MBI LAB & CARE
Mind-Body Interface Lab & Care
身心介面研究與健康中心

TSNPR

MBI Symposium
Mind-Body Interface Symposium

16th Mind-Body Interface International Symposium
2026 ISNPR Regional Meeting

Transforming Mind-Body Medicine:

From Lab to Life — AI, Neuromodulation, and Lifestyle Medicine

2026 Nov. 14-16, 2026
Taichung, Taiwan



PLENARY KEYNOTE SPEAKERS



United Kingdom



Taiwan



Taiwan



United Kingdom

INDEX

General Information

Cover 1

Chairman's Message 3

Venue 4

Floor Plan | Parking Map | Transportation

Day 1 – Scientific Program Overview 6

Day 2 – Scientific Program Overview 7

Day 3 –Technical Tour

Plenary Keynote 1 | Plenary Keynote 2 | Plenary Keynote 3 | Plenary Keynote 4

Awards | Registration | Partners | Sponsor

← Back to Index



Dear Colleagues, Distinguished Guests, and Partners,

It is my distinct honor and great pleasure to announce the **16th Mind-Body Interface International Symposium and 2026 ISNPR Regional Meeting**, to be held in the vibrant city of Taichung on **November 14–16, 2026**. Organized by the **Mind-Body Interface Research Center (MBI Lab & Care)** at China Medical University Hospital, we are proud to continue our longstanding partnership with the **International Society for Nutritional Psychiatry Research (ISNPR)** and the **Taiwanese Society for Nutritional Psychiatry Research (TSNPR).**

Building upon **our legacy of excellence,** this year's theme — **"Transforming Mind-Body Medicine: From Lab to Life — AI, Neuromodulation, and Lifestyle Medicine"** — marks a pivotal shift from theoretical discovery to actionable breakthroughs. In the rapidly evolving landscape of the 2020s, the integration of Artificial Intelligence and digital innovation has redefined the boundaries of healthcare. Our mission for this symposium is to bridge the gap **"From Lab to Life,"** accelerating the journey of translational neuroscience into the hands of those who need it most.

This year, we explore the cutting edge of Digital Intelligence as a catalyst for precision. We will delve into how Clinician-in-the-Loop AI platforms are revolutionizing interventions and how digital biomarkers are enabling objective, home-based health monitoring. From the neuro-circuitry of adolescents to the neurocognitive resilience of the elderly, we are showcasing scalable solutions that bring high-tech care into daily life.

The **16th MBI Symposium** remains the premier interdisciplinary stage for Translational Neuroscience. Our program features a comprehensive fusion of:

- **Precision Lifestyle Medicine:** Integrating the latest evidence in **Nutritional Psychiatry** and **Precision Exercise** to treat and prevent brain disorders.
- **Advanced Neuromodulation:** Exploring the next generation of non-invasive brain stimulation guided by real-time neuroimaging.
- **Bio-Psycho-Immunology:** Unraveling the molecular secrets of the gut-brain-microbe axis and metabolic processes.
- **Industry & Policy Innovation:** Highlighting the pathways for Patent Strategy, Intellectual Property (IP) development, and Academic-Industry Collaboration to turn scientific excellence into sustainable health technologies.

We cordially invite biomedical researchers, clinicians, technology innovators, and policymakers to join this inspiring dialogue. For our governmental and industry partners, this symposium offers a unique window into the future of Precision Wellbeing and a platform to foster strategic alliances that will shape the healthcare economy.

Experience the world-renowned hospitality of Taichung and join us in this transformative journey. Let us work together to harness Digital Intelligence, ensuring that every breakthrough in the lab leads to a better life for all.

We look forward to welcoming you to Taichung in 2026!

Sincerely yours,

Kuan-Pin Su, M.D., Ph.D.

Chairman, 16th MBI International Symposium

Director, Mind-Body Interface Research Center (MBI Lab & Care), CMU Hospital, Taiwan

Professor, College of Medicine, China Medical University (CMU), Taichung, Taiwan

Deputy Superintendent, An-Nan Hospital, CMU, Tainan, Taiwan

Vice President, International Society for Nutritional Psychiatry Research (ISNPR)

Founding President, Taiwanese Society for Nutritional Psychiatry Research (TSNPR)

← Back to Index

China Medical University (Shuinan Campus)

Conference Hall, B2 floor, Excellence Building China Medical University (Shuinan Campus), Taichung, Taiwan

Address: No. 100, Sec. 1, Jingmao Rd., Taichung City 406040, Taiwan

中國醫藥大學 (水湳校區) 卓越大樓 B2 國際會議廳 406040 台中市北屯區經貿路一段100號

Complete Floor Plan of Shuinan Campus



← Back to Index

Parking Map Around Campus



Transportation to the Conference Venue

From THSR Taichung Station

By Taxi: Taxi stand is at the P2 Parking lot, Exit 7. Typical fare is around NT$400 or US$13 to China Medical University (Shuinan Campus) directly.

From Kuo-Kuang Metro Transport (Shuinan Station)

By Taxi: Typical fare is around NT$130 or US$5 to China Medical University (Shuinan Campus) directly.

← Back to Index

Saturday, Nov 14 2026

Venue: Conference Hall, B2 floor, Excellence Building China Medical University (Shuinan Campus), Taichung, Taiwan.

08:30 - 09:00	**Registration**
09:00-09:20	**Opening Ceremony**
09:20- 10:00	**Plenary Keynote 1** From Lab to Life: Leveraging IMUs and Smartphones to Predict Disease Progression through Digital Mobility Outcomes **Vita Lanfranchi, Ph.D. (UK):** Professor of Digital Health and Pervasive Computing, School of Computer Science, University of Sheffield United Kingdom **Chair: Yun-Ru Lu (Taiwan)**
10:00-10:20	**Coffee break (Venue: Corridor of Conference Hall)**
10:20-12:00	**S11. Decoding the Metabolism–Brain–Sleep Axis**
12:00 – 13:00	Lunch Break, Poster Judge Session (B202, B203)
13:00 – 13:30	**5-min Poster Blitz**
13:30 – 15:10	**S12. Advances in Brain Modulation**
15:10 – 16:00	**Special Lecture [中文演講] 創新精準醫療與法規 (Innovative Precision Medicine and Regulatory Affairs)**
16:00 – 16:20	Coffee Break @Corridor of Conference Hall
16:20 – 17:00	**Plenary Keynote 2** Mind-Body Interventions and Psychoneuroimmune Modulation in Depression in the Digital Era **Kuan-Pin Su, M.D., Ph.D. (Taiwan):** Director, Mind-Body Interface Research Center (MBI-Lab & Care), China Medical University Hospital **Chair: Jen-Pang Wang (Taiwan)**
18:30 - 21:00	Welcome Dinner @Grand Ballroom I, B1, National Hotel Taichung *Additional Purchase *Shuttle Bus Provided

← Back to Index

Sunday, Nov 15 2026

Venue: Conference Hall, B2 floor, Excellence Building China Medical University (Shuinan Campus), Taichung, Taiwan.

08:15 - 08:30	**Registration**
08:30- 09:10	**Plenary Keynote 3** From COVID-19 to the Digital Pandemic: Are We Raising a Generation Under Continuous Stress? **Jane Pei-Chen Chang, M.D., M.Sc., Ph.D. (Taiwan):** Director, Child Psychiatry Division, Department of Psychiatry, China Medical University Hospital **Chair: Jui-Cheng Chen (Taiwan)**
09:10 - 10:30	**S21. Gut–Brain Axis & Psychobiotics in Psychiatry**
10:30 - 10:50	**Coffee break (Venue: Corridor of Conference Hall)**
10:50 - 12:00	**S22. Nutritional Psychiatry: Clinical Insights from Japan**
12:00 - 13:00	Lunch Break (B202, B203)
13:00 - 14:40	**S23. Rewiring the Inflamed Mind: Immuno-Psychiatry**
14:40 - 15:00	Coffee Break @Corridor of Conference Hall
15:00 - 16:00	**S24. AI in Psychiatry: From Research to Clinic**
16:00 - 16:40	**Plenary Keynote 4** PK4. Harnessing Artificial Intelligence to Improve Clinical Outcomes in Mental Health: The Role of Large Language Models and Clinician Decision Support Systems **Christoph U. Correll, M.D. (USA):** Professor, Department of Psychiatry, Zucker School of Medicine at Hofstra/Northwell **Chair: Ming-Chyi Huang (Taiwan)**
16:40 - 17:00	Award Ceremony & Closing

←Back to daily schedule



Plenary Keynote 1 - Speaker

Vita Lanfranchi, Ph.D.

Professor, School of Computer Science,
The University of Sheffield

09:00- 10:00

Abstract →Next page

"From Lab to Life: Leveraging IMUs and Smartphones to Predict Disease Progression Through Digital Mobility Outcomes"

CURRICULUM VITAE

Dr. Vitaveska Lanfranchi is a Professor of Digital Health and Pervasive Computing at the University of Sheffield, in the School of Computer Science. Her interdisciplinary research focuses on the intersection of ubiquitous computing, AI, and human behaviour monitoring, with a specific interest in remote mobility monitoring for healthcare. A specialist in participatory design, she develops intelligent systems to improve healthcare decision-making. She has extensive experience in applying and managing complex, high-impact research grants for UK and EU funding bodies. Vita has also published extensively in top-tier journals and international conferences, covering pervasive sensing, AI frameworks for disease prediction (including Alzheimer's and respiratory conditions), and the ethics and trust of digital health technologies. Previously CEO of the spin-out K-Now, she brings a unique blend of academic and industrial expertise to her work.

Research Interests: Digital Health, Co-Design, Mobility Monitoring, AI for disease progression

←Back to Plenary Keynote 1 – Speaker CV

From Lab to Life: Leveraging IMUs and Smartphones to Predict Disease Progression Through Digital Mobility Outcomes

Vita Lanfranchi, Ph.D.
Professor, School of Computer Science, The University of Sheffield

Abstract

Traditional clinical scales and assessments, such as the expanded disability severity scale (EDSS) or "no evidence of progression" (NEP) in multiple sclerosis (MS), often rely on infrequent, in-clinic functional tests. These methods are criticised for being non-linear and having poor sensitivity to subtle changes over time.
Walking mobility is a "vital sign" of functional health that can capture the multisystemic effects of disease severity in real-world settings. Digital Mobility Outcomes (DMOs) are behavioural and spatiotemporal parameters derived from wearable technology that offer a promising solution for continuous, objective remote monitoring.
This talk will discuss the utility of DMOs for predicting disease progression, specifically exploring two distinct methodologies: clinical-grade Inertial Measurement Units (IMUs) and smartphone-based applications.
Wearable IMU sensors are ideal for high-resolution, clinical-grade gait analysis and prediction of disability progression, while smartphone applications are best suited for large-scale, accessible monitoring that integrates critical environmental contextual factors like weather or pollution.

Using data from the Mobilise-D research project, we applied these methods in two different studies to demonstrate the potential of IMU and smartphone-based mobility monitoring.

1. IMU Study: 602 people with MS (pwMS) wore sensors (Axivity AX6 or McRoberts MM+) on the lower back for 7-day "free-living" periods every 6 months. Machine Learning (ML) models were trained on 24 weekly aggregate DMOs to predict EDSS and future NEP status. A complementary analysis focused on predicting "progressors"—defined as pwMS transitioning from an EDSS < = 4.0 (limited impairment) to EDSS >4.5 (observed mobility disability) over consecutive timepoints
2. Smartphone Study: 56 participants from the Technical Validation Study (including MS, COPD, and Parkinson's cohorts) carried a smartphone (Samsung Galaxy S9) for 7 days to record mobility via GPS and onboard accelerometers. This study integrated environmental contextual factors, such as temperature and wind speed, by fusing GPS trajectories with meteorological data.

Support Vector Machine (SVM) models demonstrated the highest performance (accuracy 0.77, macro-recall 0.72) in identifying pwMS with meaningful EDSS progression. These models could identify at-risk patients one year prior to a confirmed clinical change in the EDSS. Feature importance analysis highlighted that walking bout duration was a significant contributor to these predictions.
ML models were highly effective at identifying patients in advance of reaching the classic NEP target. The top-performing models for future NEP status (t+1) actually outperformed models for current status (t), with an accuracy of 0.71 and macro-recall of 0.72. DMOs related to walking speed, stride length, and stride duration were found to be the most predictive of progression. Smartphone data revealed that mobility-impaired groups respond to environmental factors in distinct patterns; for instance, the MS cohort exhibited a unique behaviour where cadence increased with temperature, unlike other groups.
DMOs serve as scalable digital biomarkers that can identify risk of progression earlier than traditional in-clinic tests. While IMUs provide the precision required for clinical-grade tracking, smartphone applications offer the scalability and contextual depth necessary to understand how environmental thresholds impact a patient's daily life and are more acceptable to patients for longitudinal studies. Together, these methods support a shift toward community-based, personalised care.

← Back to Index

S11. Decoding the Metabolism–Brain–Sleep Axis

Chair: Tae Kim (South Korea)

Time: 10:20 - 12:00 (100 min)

Time	Speakers	Title
S 11- 1	Dr. Insung Park (Japan)	Metabolic Inflexibility During Sleep: Insights From Whole-Room Indirect Calorimetry
S 11- 2	Jae Gwan Kim (South Korea)	Engineered Prefrontal fNIRS Functional Connectivity Features for Asymptomatic Alzheimer's Disease Detection
S 11- 3	Dr. Dongsung Lee (South Korea)	Epigenomic and Chromosomal Architectural Reconfiguration in Developing Human Frontal Cortex and Hippocampus
S 11- 4	Dr. Tae Kim (South Korea)	REM Sleep Suppression During Diazoxide-Induced Acute Hyperglycemia in a Non-Diabetic Mouse Model
12:00 – 13:00	Lunch Break, Poster Judge Session (B202, B203)	

←Back to Session 11 lineup



SESSION CHAIR

Prof. Piotr Gałecki,

Professor and Chair, Department of Biomedical Science and Engineering, Gwangju Institute of Science and Technology
(South Korea)

S 11 **10:20 - 12:00**

" Summary of Session"

Mind-body medicine is being reshaped by the convergence of AI, neuromodulation, and lifestyle medicine — the central themes of this symposium. Aligned with the umbrella vision of moving "From Lab to Life," this session decodes the metabolism–brain–sleep axis across four complementary scales, from lifestyle-driven physiology to single-cell molecular substrates, and from AI-enabled early detection to sleep- based intervention. The session opens with whole-body indirect calorimetry from the metabolic chamber, where minute-by-minute quantification of energy expenditure and substrate oxidation reveals how diet, light, and circadian conditions — the modifiable levers of lifestyle medicine — shape sleep architecture and 24-hour energy metabolism, establishing the mechanistic foundation linking metabolism to sleep. Building on this foundation, AI-enabled brain biomarkers come into view: functional near-infrared spectroscopy (fNIRS) combined with machine learning is shown to distinguish preclinical Alzheimer's disease from cognitively normal aging, providing a low-cost, portable, AI-driven tool for early disease detection that brings brain-health screening from the laboratory into community use. The session then descends to the molecular substrate of the brain: joint single-nucleus profiling of 3D chromatin conformation and DNA methylation (sn-m3C-seq) across more than fifty thousand cells reveals temporally distinct epigenomic and chromatin architectural reorganization of the developing human frontal cortex and hippocampus, with cell-type-specific regulatory regions that overlap putatively causal common variants for schizophrenia — defining the microscopic, neurodevelopmental molecular basis of psychiatric risk. The session closes by returning from molecule to lifestyle intervention: hyperglycemia is shown to suppress and fragment REM sleep through dietary and dopaminergic mechanisms in preclinical models, positioning sleep-based interventions — alongside glycemic control — as actionable, lifestyle-medicine levers for improving metabolic and cognitive outcomes. Together, the four talks chart a coherent translational arc — lifestyle-metabolic measurement, AI- enabled brain biomarkers, microscopic brain regulomes, and sleep- based therapeutic intervention — and embody the symposium's vision of transforming mind-body medicine through AI, neuromodulation, and lifestyle medicine, From Lab to Life.

←Back to Session 11 lineup



SPEAKER

Dr. Insung Park

Associate Professor, Institute of Global Well-Being Science, Hirosaki University
(Japan)

S 11-1

Abstract → Next

"Metabolic Inflexibility During Sleep: Insights From Whole-Room Indirect Calorimetry"

CURRICULUM VITAE

←Back to Speaker CV

Metabolic Inflexibility During Sleep: Insights From Whole-Room Indirect Calorimetry

Dr. Insung Park

Associate Professor, Institute of Global Well-Being Science, Hirosaki University

Abstract

Metabolic flexibility—the capacity to shift fuel oxidation in accordance with nutritional and physiological state—is increasingly recognized as a barometer of metabolic health. When this capacity is impaired, the resulting metabolic inflexibility is closely tied to the development of obesity and type 2 diabetes mellitus. Sleep, for most people a monophasic and thus prolonged fast, has long been assumed to drive a steady progression from carbohydrate to fat oxidation, observable as a sustained fall in the respiratory quotient (RQ). That assumption, however, does not hold. Across 127 whole-room indirect calorimetry recordings, RQ consistently re-ascended before awakening—pointing to a resumption of carbohydrate oxidation in the middle of the night, well before the fast was broken. The timing of this RQ nadir differed by sex: 3.41 ± 0.19 h after bedtime in men, but only 1.92 ± 0.30 h in women. Stratifying young non-obese men by their 24-h RQ range (zenith-to-nadir: 0.141 ± 0.002 in flexible vs. 0.104 ± 0.002 in inflexible individuals) brought the sleep phase into sharper focus. Daytime postprandial RQ responses were indistinguishable between groups; the divergence emerged only at night, when the expected RQ decline was blunted in metabolically inflexible subjects. A comparable attenuation appeared with just a 10-year age difference, and inter-individual RQ variability was substantially wider during sleep than during waking hours. These findings suggest that sleep-phase substrate metabolism can be detected as a sensitive marker of early-stage metabolic inflexibility, offering a framework for its assessment beyond conventional daytime metrics.

←Back to Session 11 lineup



SPEAKER

Dr. Jae Gwan Kim

Professor, Department of Biomedical Science and Engineering, Gwangju Institute of Science and Technology
(South Korea)

S 11-2

Abstract → Next

"Engineered Prefrontal fNIRS Functional Connectivity Features for Asymptomatic Alzheimer's Disease Detection"

CURRICULUM VITAE

←Back to Speaker CV

Metabolic Inflexibility During Sleep: Insights From Whole-Room Indirect Calorimetry

Dr. Jae Gwan Kim

Professor, Department of Biomedical Science and Engineering, Gwangju Institute of Science and Technology

Abstract

Background:

Alzheimer's disease (AD) pathology begins 20–30 years before symptoms, making the asymptomatic stage the critical window for intervention. Yet current diagnostic tools—amyloid PET, CSF, and MRI—are invasive, costly, and unsuitable for population-scale screening. Functional near-infrared spectroscopy (fNIRS) offers a portable, low-cost, non-invasive alternative for measuring prefrontal functional connectivity (FC), a promising early biomarker.

Methods:

We analyzed 156 participants (81 healthy controls; 75 cognitively normal, amyloid-positive individuals) from the GARD cohort, using paired resting- state and verbal fluency task recordings from a portable 4-channel prefrontal fNIRS device. From HbO-based Fisher-z FC across four channel pairs, we engineered three biologically motivated features capturing distinct phases of the prefrontal network response: Task Activation Delta (Δ1), Task Transition Delta (Δ2), and Task Endurance Slope (β). Group differences were assessed with Mann- Whitney U tests, Cohen's d, ANCOVA, and FDR correction, followed by classification using nine machine-learning models.

Results:

Eleven of 18 features survived FDR correction, with interhemispheric pairs ch1–ch3 and ch1–ch4 consistently most discriminative; all findings were independent of sex. Among nine classifiers, SVM-RBF performed best (accuracy 0.699, sensitivity 0.747, AUC 0.706), and engineered features substantially outperformed raw FC baselines. SHAP analysis confirmed biological validity, with statistical and machine-learning results closely aligned.

Conclusion:

A portable, low-cost 4-channel fNIRS device can detect subtle prefrontal network disruptions in asymptomatic amyloid-positive individuals, supporting a pathway toward accessible early AD screening.

←Back to Session 11 lineup



SPEAKER

Dr. Dongsung Lee

Associate Professor, Department of Biomedical Sciences, Seoul National University College of Medicine
(South Korea)

S 11-3 Abstract → Next

"Epigenomic and Chromosomal Architectural Reconfiguration in Developing Human Frontal Cortex and Hippocampus"

CURRICULUM VITAE

←Back to Speaker CV

Epigenomic and Chromosomal Architectural Reconfiguration in Developing Human Frontal Cortex and Hippocampus

Dr. Dongsung Lee

Associate Professor, Department of Biomedical Sciences, Seoul National University College of Medicine

Abstract

The human frontal cortex and hippocampus play critical roles in learning and cognition. We investigated the epigenomic and 3D chromatin conformational reorganization during the development of the frontal cortex and hippocampus, using more than 53,000 joint single-nucleus profiles of chromatin conformation and DNA methylation (sn-m3C-seq). The remodeling of DNA methylation predominantly occurs during late-gestational to early-infant development and is temporally separated from chromatin conformation dynamics. Neurons have a unique Domain-Dominant chromatin conformation that is different from the Compartment-Dominant conformation of glial cells and non-brain tissues. We reconstructed the regulatory programs of cell-type differentiation and found putatively causal common variants for schizophrenia strongly overlap with chromatin loop-connected, cell-type-specific regulatory regions. Our data demonstrate that single-cell 3D-regulome is an effective approach for dissecting neuropsychiatric risk loci.

←Back to Session 11 lineup



SPEAKER

Dr. Tae Kim

Professor and Chair, Department of Biomedical Science and Engineering, Gwangju Institute of Science and Technology
(South Korea)

S 11-4

Abstract → Next

"REM Sleep Suppression During Diazoxide-Induced Acute Hyperglycemia in a Non-Diabetic Mouse Model"

CURRICULUM VITAE

Dr. Tae Kim is Professor and Chair of the Department of Biomedical Science and Engineering at the Gwangju Institute of Science and Technology (GIST), and Founder & CEO of TEDi MEDi. He is a Licensed Psychiatrist and Licensed Physician under the Ministry of Health & Welfare, South Korea, receiving his respective licenses in 2004 and 1999. He received the Lily Research Award from the Korean Neuropsychiatric Association in 2016 and the Lundbeck Neuroscience recognition. He serves as Vice President of the Korean Society of Sleep Medicine (2026–2027), Director of Basic Research at the Korean Sleep Research Society (2024–2026), and Associate Editor of the Journal of Alzheimer's Disease (2024–present). His research publications cover dopaminergic dysregulation, REM sleep fragmentation, ADHD-like behaviors, microbiome-derived metabolites, amyloidopathy, transcranial stimulation, and cortical gamma-band oscillations. His publication record includes articles in Psychiatry Research (2023), Brain, Behavior, and Immunity (2024), Molecular Psychiatry (2024), Translational Neurodegeneration (2021), and PNAS (2015). His total number of SCI(E) papers is 56+.

←Back to Speaker CV

REM Sleep Suppression During Diazoxide-Induced Acute Hyperglycemia in a Non-Diabetic Mouse Model

Dr. Tae Kim

Professor and Chair, Department of Biomedical Science and Engineering, Gwangju Institute of Science and Technology

Abstract

Sleep is essential for metabolic homeostasis, and disruption of sleep or circadian rhythms impairs glucose control and hormonal balance. Sleep disturbances are frequent in diabetes and often attributed to nocturia or neuropathic symptoms, yet the recurrent coexistence of hyperglycemia and poor sleep suggests a direct metabolic effect on sleep regulation. The acute impact of hyperglycemia on sleep architecture remains unclear. Acute hyperglycemia was induced in C57BL/6 mice by 8-hour fasting followed by intraperitoneal diazoxide injection and oral glucose gavage, producing ~6 h of elevated blood glucose. EEG/EMG recordings were acquired during this period, with DMSO- and saline-treated mice as controls. For immunohistochemistry, mice (n = 3/group) were sacrificed 3 h after treatment, perfused with PBS and 4% paraformaldehyde, cryoprotected, sectioned (40 μm), and stained with anti-NeuN and anti-c-Fos antibodies. c-Fos$^+$/NeuN$^+$ neurons were quantified. To monitor locus coeruleus (LC) activity, a TH-Cre mouse received AAV(DJ)-EF1a-DIO-GCaMP6f injection into the right LC and optic fiber implantation 0.15 mm above the site, followed by simultaneous EEG/EMG and fiber photometry. Acute hyperglycemia markedly suppressed REM sleep, reducing REM percentage, bout length, and duration. EEG absolute power decreased across wake and NREM, indicating dampened cortical activity. LC fiber photometry and c- Fos staining showed increased noradrenergic LC activity during NREM after diazoxide. Acute hyperglycemia directly suppresses REM sleep and cortical activation, with LC hyperactivity mediating REM inhibition. This work provides the first experimental demonstration that transient glucose elevation alone rapidly disrupts REM sleep through LC activation, revealing a metabolic–neuromodulatory mechanism linking hyperglycemia to impaired sleep regulation.

←Back to daily schedule

5-min Poster Blitz

Time: 13:00 - 13:30 (30min)

Time	Speakers	Title
PP003	**Szu-Han Chen (Taiwan)**	Effects of Omega-3 Polyunsaturated Fatty Acids on Metabolic Parameters in Patients Treated With Atypical Antipsychotics: An Exploratory Meta-analysis of Randomized Controlled Trials
OP004	**Sanae Hasegawa-Ishii (Japan)**	Nasal, but not Gut, Microbiota are Associated With Hippocampal Neuroimmune Alterations in a Mouse Model of Eosinophilic
5PB007	**Jianjung Ying (Taiwan)**	Blunted Nocturnal HRV Recovery, Reduced Physical Activity, and Lower Sleep Quality in Major Depressive Disorder: A Digital Phenotyping Study Using Wearable Devices in a Taiwanese Cohort
5PB006	**Aimal Khan (Taiwan)**	PUFA-derived Lipid Mediator and Endocannabinoid Signaling Pathways in Major Depressive Disorder: A Mendelian Randomization Study
5PB013	**Wen-Chin Yu (Taiwan)**	Food Consumption by Degree of Processing and Emotional Distress Among Taiwanese Adolescents and Adults

13:30 – 15:10 S12. Advances in Brain Modulation

← Back to Index

S12. Advances in Brain Modulation

Chair: Jia-Jin Chen (Taiwan)

Time: 13:30 - 15:10 (100 min)

Time	Speakers	Title
S 12- 1	Chien-An Chen, Ph.D. Candidate (Taiwan)	Closed-Loop Neuromodulation Using fNIRS-Guided HD-tES: Opportunities and Future Directions
S 12- 2	Dr. Che-Wei Lin (Taiwan)	Connectivity-Aware Deep Learning on Wearable fNIRS: Subject-Independent Anxiety Assessment and Its Path to Closed-Loop Neuromodulation
S 12- 3	Dr. Tsung-Hsun Hsieh (Taiwan)	Neuromodulation Strategies in Neurorehabilitation: From Mechanistic Insights to Precision Neurotherapy
S 12- 4	Dr. Yi-Jing Huang (Taiwan)	Neurophysiological Profiling and High-Definition Transcranial Electrical Stimulation Treatment for Generalized Anxiety Disorder
15:10 - 16:00	Special Lecture [中文演講] 創新精準醫療與法規 (Innovative Precision Medicine and Regulatory Affairs)	

←Back to Session 12 lineup



SESSION CHAIR

Jia-Jin Chen (陳家進), Ph.D.
Distinguished Professor, Department of Biomedical Engineering, National Cheng Kung University
(Taiwan)

S 12

13:30 - 15:10

S12: Emerging Neurotechnologies for Brain Modulation

" Summary of Session"

This session aims to highlight the emerging transition of neuromodulation from conventional, protocol-based stimulation toward mechanism-informed, biomarker-guided, and personalized neurotherapy. By integrating perspectives from neuroengineering, artificial intelligence, neurophysiology, rehabilitation, and psychiatry, the session will address how individualized neural states can be characterized and ultimately incorporated into adaptive and closed-loop neuromodulation strategies. The first presentation will review the concept and technological framework of closed-loop neuromodulation using functional near-infrared spectroscopy (fNIRS)- guided high-definition transcranial electrical stimulation (HD-tES), emphasizing individualized targeting, real-time cortical monitoring, and adaptive stimulation. Extension of this framework from sensing to intelligent interpretation will presenting connectivity-aware deep learning of wearable fNIRS signals for subject- independent anxiety assessment and its potential as an interpretable biomarker for closed-loop control. Addition research works from CGU Hospital will provide a broader mechanistic and translational perspective on neuromodulation in neurorehabilitation, covering electrical, magnetic, ultrasound, and gamma- frequency sensory stimulation and their effects on neural excitability, oscillations, and plasticity. Finally, research outcome from NTU hospital will bridge these concepts to clinical application by presenting neurophysiological profiling and HD- tES treatment in generalized anxiety disorder, with particular emphasis on prefrontal dysfunction, brain–heart interactions, and treatment-related autonomic changes. Collectively, this session maps the current landscape and future trajectory of neuromodulation, integrating advances in mechanism, sensing, biomarkers, personalized stimulation, and clinical translation to provide a broad perspective on the development of precision and closed-loop neurotherapy.

←Back to Session 12 lineup



SPEAKER

Chien-An Chen, Ph.D. Candidate

PhD Candidate, Department of Biomedical Engineering, National Cheng Kung University
CEO, WeiAn Medical Technologies Co., Ltd.
(Taiwan)

S 12-1 Abstract → Next

"Closed-Loop Neuromodulation Using fNIRS-Guided HD-tES: Opportunities and Future Directions"

CURRICULUM VITAE

←Back to Speaker CV

Closed-Loop Neuromodulation Using fNIRS-Guided HD-tES: Opportunities and Future Directions

Chien-An Chen, Ph.D. Candidate

PhD Candidate, Department of Biomedical Engineering, National Cheng Kung University
CEO, WeiAn Medical Technologies Co., Ltd.

Abstract

Background:
Non-invasive brain stimulation has emerged as a promising approach for modulating neural activity and treating neurological and psychiatric disorders. However, substantial inter- and intra-individual variability, uncertain dose–response relationships, and the lack of standardized biomarkers for target engagement remain major limitations. Closed-loop neuromodulation, in which stimulation is dynamically guided by neurophysiological feedback, may provide a path toward more individualized and effective interventions.

Methods: This work reviews recent developments in non-invasive neuromodulation and proposes a closed-loop framework integrating high-definition transcranial electrical stimulation (HD-tES) with functional near-infrared spectroscopy (fNIRS). The framework consists of individualized spatial targeting based on neuroimaging, stimulation dose and waveform optimization, and real-time fNIRS monitoring of cortical hemodynamic responses. A multi-channel HD-tES platform and a wearable fNIRS prototype were developed as the hardware basis for future system integration.

Results: Current evidence demonstrates increasing adoption of connectivity- and imaging- guided stimulation targets across stroke, cognitive, and psychiatric applications. Nevertheless, stimulation parameters and dose–response relationships remain heterogeneous. Our proposed architecture combines personalized targeting, flexible HD-tES waveforms, and real-time cortical hemodynamic monitoring, providing the technical foundation for adaptive stimulation based on individual brain responses.
Conclusion: The integration of HD-tES and fNIRS represents a feasible strategy toward personalized closed-loop neuromodulation. Future work will focus on simultaneous fNIRS–HD-tES measurement, real-time biomarker extraction, and adaptive optimization of stimulation parameters to improve treatment precision and reproducibility.

←Back to Session 12 lineup



SPEAKER

Dr. Che-Wei Lin

Professor, Department of Biomedical Engineering,
National Cheng Kung University
(Taiwan)

S 12-2

Abstract → Next

"Connectivity-Aware Deep Learning on Wearable fNIRS: Subject-Independent Anxiety Assessment and Its Path to Closed-Loop Neuromodulation"

CURRICULUM VITAE

Dr. Che-Wei Lin is a Professor in the Department of Biomedical Engineering at National Cheng Kung University, Tainan, Taiwan, where he has served since October 2025. He holds joint appointments at the Medical Device Innovation Center, Institute of Gerontology, Institute of Medical Informatics, and Department of Biotechnology and Bioindustry Sciences. Since August 2026, he has also served as Director of the Startup Accelerator Center at National Cheng Kung University. Previously, he was an Associate Professor (2021–2025) and Assistant Professor (2016–2021) in the Department of Biomedical Engineering at the same university. From 2011 to 2016, he worked as a Postdoctoral Researcher and Assistant Researcher in the Department of Electrical Engineering. His research achievements have been recognized through several awards, including the 2026 Excellent Industrial Potential Award at SEMBA, the 2025 Asia-Pacific Research Networking Fellow recognition from the International Federation for Medical and Biological Engineering (IFMBE), the 2025 Rising Star Research Award from the College of Engineering at National Cheng Kung University, the 2024 IFMBE Young Investigator Competition Second Place Award, and the Taiwan National Innovation Award in 2023 and 2024. His research publications focus on biomedical engineering, artificial intelligence, ECG-based arrhythmia detection, deep learning, and medical technology innovation.

←Back to Speaker CV

Connectivity-Aware Deep Learning on Wearable fNIRS: Subject-Independent Anxiety Assessment and Its Path to Closed-Loop Neuromodulation

Dr. Che-Wei Lin

Professor, Department of Biomedical Engineering, National Cheng Kung University

Abstract

Background:

Generalized anxiety disorder is diagnosed largely through subjective interviews and self-report scales, leaving a need for objective, low-burden physiological markers. Functional near-infrared spectroscopy (fNIRS) offers portable, motion-tolerant measurement of prefrontal hemodynamics, but conventional channel-wise analyses discard the inter-channel connectivity that characterizes anxious brain states.

Method: Sixty-two participants (anxiety and healthy controls) completed four cognitive tasks — Go/No-Go, 1-back working memory, verbal fluency, and serial subtraction — during prefrontal fNIRS recording. Signals were preprocessed in HOMER3 with wavelet-based motion-artifact correction, reducing artifact spikes by 23% overall and 31% in the anxiety group. Each trial was represented as a graph in which channels are nodes and edges connect channel pairs with absolute Pearson correlation above 0.1, weighted by correlation and coherence. Two graph attention networks were compared: a spatial model (GATv2) and a spatial-temporal model applying GATv2 to 1.6-second windows aggregated by a gated recurrent unit with additive attention. A 3D vision-transformer grid encoding served as a non-graph baseline. Models were evaluated using 5-fold, 10-fold, and leave-one-subject-out (LOSO) cross-validation.

Result: The spatial-temporal graph model reached 0.77–0.78 accuracy under 10- fold validation and 0.82 accuracy with 0.84 F1 under LOSO, consistently exceeding the spatial-only and grid-based baselines. Explainable-AI channel analysis localized discriminative connectivity, and anxious participants showed lower task-evoked activation than controls. Conclusion: Connectivity-aware deep learning on wearable fNIRS provides a subject-independent, interpretable marker of anxiety, supporting objective assessment and future closed-loop neuromodulation.

←Back to Session 12 lineup



SPEAKER

Dr. Tsung-Hsun Hsieh

Professor, School of Physical Therapy and Graduate Institute of Rehabilitation Science, College of Medicine, Chang Gung University
Director, The Center for Innovation and Incubation, Chang Gung University
(Taiwan)

S 12-3

Abstract → Next

"Neuromodulation Strategies in Neurorehabilitation: From Mechanistic Insights to Precision Neurotherapy"

CURRICULUM VITAE

Dr. Tsung-Hsun Hsieh is a Professor at the School of Physical Therapy and Graduate Institute of Rehabilitation Science, College of Medicine, Chang Gung University, Taiwan. He also serves as Director of the Center for Innovation and Incubation at Chang Gung University, a position he has held since 2020. From 2010 to 2012, he was a Research Scholar and Postdoctoral Research Fellow in the Department of Neurology at Boston Children's Hospital and Harvard Medical School, USA. His research has received several recognitions, including Bronze (2020), Silver (2022), and Gold (2024) Medals at the Taiwan Innotech Expo Invention Competition, as well as the Editor's Choice Award from Brain Sciences in 2020. His research focuses on neuroscience, neurorehabilitation, Parkinson's disease, cortical electrical stimulation, and neuromodulation. His selected publications include studies on the neuroprotective effects of cortical electrical stimulation, temporal interference stimulation, and transcranial direct current stimulation in Parkinson's disease.

←Back to Speaker CV

Neuromodulation Strategies in Neurorehabilitation: From Mechanistic Insights to Precision Neurotherapy

Dr. Tsung-Hsun Hsieh

Professor, School of Physical Therapy and Graduate Institute of Rehabilitation Science, College of Medicine, Chang Gung University
Director, The Center for Innovation and Incubation, Chang Gung University

Abstract

Background:

Neuromodulation is an emerging therapeutic strategy in neurorehabilitation that can modulate neural activity, promote neuroplasticity, and facilitate functional recovery. This presentation provides an integrated overview of neuromodulation strategies, from mechanistic insights to clinical translation and precision neurotherapy.

Method: Mechanistic and translational evidence from experimental and clinical studies of repetitive transcranial magnetic stimulation (rTMS), transcranial direct current stimulation (tDCS), cortical electrical stimulation (CES), and transcranial focused ultrasound (tFUS) will be reviewed. Electrophysiological measures, including electroencephalography (EEG) and motor evoked potentials (MEP), will be discussed as tools for characterizing neuromodulatory responses. Emerging gamma-frequency sensory stimulation using rhythmic light and sound will also be introduced.

Result:

Current evidence demonstrates that neuromodulation can alter cortical excitability, neural oscillations, synaptic plasticity, and network-level activity, supporting its therapeutic potential across neurological disorders. Gamma- frequency sensory stimulation, particularly around 40 Hz, can entrain neural oscillations and may provide a novel approach for modulating pathological brain activity in neurodegenerative conditions. However, treatment responses remain variable across individuals.

Conclusion: Neuromodulation is evolving toward mechanism-driven and personalized intervention. Integrating neurophysiological biomarkers, individualized stimulation parameters, and emerging approaches such as gamma-frequency sensory stimulation may facilitate the development of precision neurotherapy and expand future applications in neurorehabilitation.

←Back to Session 12 lineup



SPEAKER

Dr. Yi-Jing Huang

Assistant professor, School of Occupational Therapy, College of Medicine, National Taiwan University Adjunct occupational therapist, Department of Physical Medicine and Rehabilitation, National Taiwan University Hospital **(Taiwan)**

S 12-4

Abstract → Next

"Neurophysiological Profiling and High-Definition Transcranial Electrical Stimulation Treatment for Generalized Anxiety Disorder"

CURRICULUM VITAE

Dr. Yi-Jing Huang is an Assistant Professor at the School of Occupational Therapy, College of Medicine, National Taiwan University, and an Adjunct Occupational Therapist in the Department of Physical Medicine and Rehabilitation at National Taiwan University Hospital. Previously, she served as an Adjunct Assistant Professor at National Taiwan University (2020–2023) and worked as a Postdoctoral Researcher at Taipei Medical University (2020–2021) and the University of Sydney, Australia (2018–2019). She received her Ph.D. in Occupational Therapy (2018) and B.S. in Occupational Therapy (2011) from National Taiwan University. In 2023, she received the 20th National Innovation Award – Academic Research Innovation Award. Her research focuses on occupational therapy, cognitive and emotional processing, prefrontal brain activity, and anxiety disorders. Her recent publications investigate prefrontal cortical activity and neural efficiency during emotional response inhibition, particularly in patients with generalized anxiety disorder and across different age groups.

←Back to Speaker CV

Neurophysiological Profiling and High-Definition Transcranial Electrical Stimulation Treatment for Generalized Anxiety Disorder

Dr. Yi-Jing Huang

Assistant professor, School of Occupational Therapy, College of Medicine, National Taiwan University
Adjunct occupational therapist, Department of Physical Medicine and Rehabilitation, National Taiwan University Hospital

Abstract

Background:

Generalized anxiety disorder (GAD) features impaired cognitive inhibition and central-autonomic dysregulation. The dynamic brain-heart interplay under cognitive load and its neuromodulation potential require investigation. This two-phase translational study investigated central-autonomic mechanisms in GAD and evaluated the clinical efficacy of high-definition transcranial electrical stimulation (HD-tES). Method: Phase 1 compared prefrontal activity (via functional near-infrared spectroscopy) and post-task heart rate variability (HRV) between GAD patients and healthy controls (HC) during an emotional Go/No-Go task. Phase 2 comprised a pilot triple-blind randomized controlled trial where 10 GAD patients received 10 sessions of active (n=7) or sham (n=3) HD-tES, targeting dorsolateral prefrontal cortex. Clinical symptoms and HRV were assessed longitudinally.

Result:

In Phase 1, GAD showed a larger accuracy decline under emotional load (87.7% to 47.2%) and unmodulated prefrontal responses compared to HC. Post-task, GAD exhibited higher LF/HF ratios ($p=.045$) and lower RMSSD ($p=.043$). Within GAD, prefrontal activation positively correlated with SDNN ($\beta=.391$) and LF/HF ($\beta=.481$). In Phase 2, active HD-tES reduced HAM-A (25.0 to 19.1), STAI-S (50.3 to 47.0), and PSQI (15.3 to 13.1), whereas sham showed no improvements. Unlike static sham responses, active LF/HF ratios displayed pronounced fluctuations, featuring a transient mid-treatment surge (peak 5.30) before progressively descending (1.90).

Conclusion:

GAD presents prefrontal inflexibility and maladaptive brain-heart interplay, where cortical engagement correlates with subsequent autonomic imbalance. Preliminary data indicate HD-tES could ameliorate anxiety and sleep symptoms. Furthermore, the dynamic LF/HF trajectory during HD-tES suggests an active autonomic reconfiguration, highlighting its potential to modulate aberrant central-autonomic pathways.

← Back to Index

Special Lecture [中文演講]創新精準醫療與法規]

(Innovative Precision Medicine and Regulatory Affairs)

Chair: Hong-Wei Yen (Taiwan)

Dean, College of Engineering, Tunghai University

Time: 15:10 - 16:00 (50 min)

Time	Speakers	Title
SL- 1	Dr. Samuel Shun-Min Wang (王舜民) (Taiwan)	**中文演講 Mandarin** 智慧醫療軟體（SaMD）與AI/ML醫材的全球法規策略 Global Regulatory Strategies for SaMD and AI/ML Medical Devices
SL- 2	Dr. Chun-Hui Liao (廖俊惠) (Taiwan)	**中文演講 Mandarin** 整合素標靶細胞外囊泡遞送 BDNF mRNA 以恢復神經免疫恒定並治療重度憂鬱症 Integrin-Targeted Extracellular Vesicles Delivering BDNF mRNA Restore Neuroimmune Homeostasis for Major Depressive Disorder Treatment
16:00 - 16:20	Coffee Break	@Corridor of Conference Hall

←Back to special lecture lineup



SPECIAL LECTURE -1- SPEAKER

Shun-Min Wang (王舜民), Ph.D.

General Manager, GEM Consultants International CORP.
(Taiwan)

SL-1

Abstract → Next

"Global Regulatory Strategies for SaMD and AI/ML Medical Device"

智慧醫療軟體（SaMD）與AI/ML醫材的全球法規策略

中文演講 Mandarin

CURRICULUM VITAE

Samuel Wang holds a Ph.D. in Biomedical Engineering and boasts over two decades of career experience in the medical devices industry. He spent 18 years at Underwriters Laboratory (UL), where he held positions such as Engineer, Regional Lead Reviewer, and Regional Operations Manager for Asia Pacific. During his tenure, Samuel successfully established an exceptional team for medical device testing and certification, and himself also qualified as Technical File assessor/reviewer for CE MDD certifications, served as 3rd party reviewer for US FDA 510K submissions, and designated as Technical Assessor/Lab Auditor for IECEE. While completing his Ph.D. dissertation, Samuel joined ITRI as the Director of Regulatory Affairs Offices in the Biomedical Technology and Device Research Laboratories. ITRI, a renowned global technology R&D institution, focuses on pioneering a better future for society. In this role, Samuel led a team responsible for assisting internal and external customers in leveraging medical regulatory science for strategic product realization processes. His responsibilities encompassed diverse projects, including those involving AI SaMD and Digital Health. Currently, Samuel works as consultant for several medical startups and serves as a speaker/trainer for companies, universities, and workshops sponsored by competent authorities.

←Back to Speaker CV

Global Regulatory Strategies for SaMD and AI/ML Medical Device
智慧醫療軟體（SaMD）與 AI/ML 醫材的全球法規策略

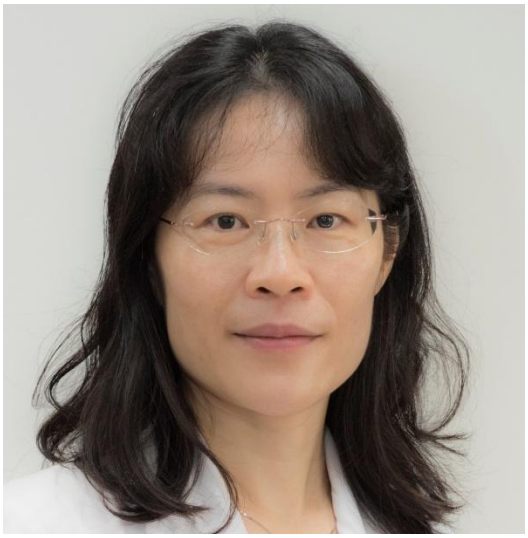
Shun-Min Wang (王舜民), Ph.D.
General Manager, GEM Consultants International CORP.

Abstract

Software as a Medical Device (SaMD) and artificial intelligence/machine learning (AI/ML)-enabled medical devices are rapidly transforming healthcare by supporting disease detection, diagnosis, risk prediction, patient monitoring, personalized treatment, and clinical workflow improvement. Unlike conventional medical devices, these products are strongly influenced by software updates, evolving datasets, model retraining, and changes in real-world clinical environments. As a result, developers face increasing challenges in demonstrating safety, effectiveness, transparency, and sustained performance throughout the product lifecycle. This presentation provides an overview of global regulatory strategies for SaMD and AI/ML medical devices. It compares key regulatory principles and market- entry considerations in the United States, European Union, and selected Asia-Pacific jurisdictions. Major topics include software qualification, risk classification, regulatory pathway selection, intended-use definition, technical documentation, clinical evaluation, and the use of international standards to support submissions across multiple markets. The presentation will also address evidence requirements specific to AI-enabled medical devices, including data quality, algorithm performance, external validation, generalizability, subgroup performance, bias, usability, human–AI interaction, and cybersecurity. Emerging approaches to software modification and algorithm change management, including predetermined change plans, model governance, version control, and post-market performance monitoring, will also be discussed. An effective global regulatory strategy should be established early and aligned with clinical development, quality management, software engineering, data governance, cybersecurity, and commercial planning. A total product lifecycle approach can reduce duplicated work, support responsible product updates, and help manufacturers deliver safe, effective, and internationally accessible AI-enabled medical technologies.

軟體醫療器材（Software as a Medical Device, SaMD）及人工智慧／機器學習 （AI/ML）醫療器材，已逐步應用於疾病偵測、診斷輔助、風險預測、病人監 測及臨床流程優化。然而，軟體版本更新、資料變動、模型再訓練，以及不 同臨床場域間的差異，也為產品查驗登記與上市後管理帶來新的挑戰。 本演講將介紹美國、歐盟、及其他主要市場之法規管理架構與全球布局策略，內容涵蓋產品屬性認定、風險分級、上市審查途徑、預期用途界定、技術文件準備及臨床評估。同時說明AI醫療器材於資料品質、演算法效能、外部 驗證、偏差、不同族群表現、泛化能力、人機互動及資安等面向的證據要求，並探討軟體變更、模型治理、版本管控與上市後效能監測。全球法規策略應於產品開發初期即納入規劃，並整合法規、臨床、品質管理、軟體工程、資料治理及商業布局，以減少重複作業，支持產品持續改版，並確保AI醫療器材之安全、效能及國際市場可行性。

←Back to special lecture lineup



SPECIAL LECTURE – 2 - SPEAKER

Chun-Hui Liao (廖俊惠), M.D.

Director of the Department of Psychiatry, China Medical University Hospital
Assistant Professor, School of Medicine, China Medical University
(Taiwan)

SL-2

Abstract → Next

"Integrin-Targeted Extracellular Vesicles Delivering BDNF mRNA Restore Neuroimmune Homeostasis for Major Depressive Disorder Treatment"

整合素標靶細胞外囊泡遞送BDNF mRNA以恢復神經免疫恆定並治療重度憂鬱症

中文演講 Mandarin

CURRICULUM VITAE

Chun-Hui Liao (廖俊惠), M.D., is the Director of the Department of Psychiatry at China Medical University Hospital, Taiwan, a position he has held since July 2023, and an Assistant Professor at the School of Medicine, China Medical University, since August 2011. He has extensive clinical experience in inpatient and outpatient psychiatric care, with expertise in schizophrenia, depression, bipolar disorder, anxiety disorders, and insomnia. His clinical interests also include geriatric psychiatry, particularly the assessment and treatment of dementia, late-life depression, and anxiety disorders through multidisciplinary care. In addition, he has experience in addiction treatment and forensic psychiatric evaluation. His selected publications cover the relationship between stroke and gastroesophageal reflux disease, anxiety and depression among women with cancer, stroke and suicide risk in individuals with severe mental illness, menstrual distress, and psychiatric outcomes among intensive care unit survivors.

←Back to Speaker CV

Integrin-Targeted Extracellular Vesicles Delivering BDNF mRNA Restore Neuroimmune Homeostasis for Major Depressive Disorder Treatment

整合素標靶細胞外囊泡遞送BDNF mRNA以恢復神經免疫恆定並治療重度憂鬱症

Chun-Hui Liao (廖俊惠), M.D.

Director of the Department of Psychiatry, China Medical University Hospital
Assistant Professor, School of Medicine, China Medical University

Abstract

Major depressive disorder (MDD) is increasingly recognized as a complex neuropsychiatric condition involving persistent neuroinflammation, impaired neuroplasticity, mitochondrial dysfunction, and oxidative neuronal injury. These interconnected pathological processes may contribute to the delayed therapeutic response and limited efficacy of conventional antidepressants, particularly in patients with treatment-resistant depression. To simultaneously enhance brain- targeted delivery and restore neurotrophic signaling, integrin-targeted extracellular vesicles (αITG-EVs) were engineered to display an integrin-specific single-chain variable fragment and carry brain-derived neurotrophic factor mRNA (mBDNF). The resulting mBDNF@αITG-EVs retained the characteristic morphology and physicochemical properties of EVs while integrating targeted cellular recognition with functional mRNA delivery. Their therapeutic activity was evaluated using a chronic stress-induced neuron–microglia–astrocyte co-culture model that recapitulated key features of the neuroinflammatory microenvironment associated with MDD. mBDNF@αITG-EVs were efficiently internalized across neuronal and glial cell populations and induced coordinated neuroimmune regulation. Treatment shifted microglia toward an anti- inflammatory phenotype, reduced the production of TNF-α and IL-1β, and increased the expression of IL-4 and IL-10. These immunomodulatory effects were accompanied by preservation of neuronal morphology and synaptic integrity. Mechanistically, mBDNF@αITG-EVs enhanced neurotrophic signaling, stabilized mitochondrial function, and attenuated oxidative stress and lipid peroxidation. Restoration of glutathione peroxidase 4 expression further indicated suppression of ferroptosis-related neuronal injury. These findings demonstrate that targeted EV-mediated delivery of BDNF mRNA can concurrently modulate neuroinflammation, neuronal plasticity, and oxidative injury, supporting its potential as a cell-free therapeutic strategy for MDD.

重度憂鬱症（major depressive disorder, MDD）日益被認為是一種複雜的神經 精神疾病，其病理機轉涉及持續性的神經發炎（neuroinflammation）、神經 可塑性受損、粒線體功能障礙，以及氧化性神經元損傷。這些彼此相互關聯 的病理過程，可能共同導致傳統抗憂鬱藥物治療反應延遲及療效受限，尤其 是在治療抗拒型憂鬱症（treatment-resistant depression）患者中更為明顯。 為了同時提升腦部標靶遞送效率並恢復神經滋養訊號，本研究設計了整合素 標靶之外泌性囊泡（integrin-targeted extracellular vesicles, αITG-EVs），使其表 面表現具有整合素專一性的單鏈可變片段（single-chain variable fragment）， 並攜帶腦源性神經滋養因子（brain-derived neurotrophic factor, BDNF）mRNA（mBDNF）。所製備的mBDNF@αITG-EVs保有細胞外囊泡 （extracellular vesicles, EVs）典型的形態及理化特性，同時結合了標靶性細胞 辨識能力與功能性 mRNA 遞送作用。 本研究利用慢性壓力誘導之神經元－小膠質細胞－星狀膠質細胞共培養模型，評估其治療活性。此模型可模擬與 MDD 相關神經發炎微環境的主要特徵。mBDNF@αITG-EVs可被神經元及各類膠質細胞有效攝取，並誘發協調性的 神經免疫調節作用。 治療後，小膠質細胞轉向抗發炎表型，腫瘤壞死因子 α（TNF-α）及介白素-1β（IL-1β）的產生下降，同時介白素-4（IL-4）與介白素-10（IL-10）的表現增加 。這些免疫調節作用並伴隨神經元形態及突觸完整性的維持。 在作用機轉方面，mBDNF@αITG-EVs 可增強神經滋養訊號傳遞、穩定粒線體 功能，並減輕氧化壓力及脂質過氧化。穀胱甘肽過氧化酶 4（glutathione peroxidase 4, GPX4）表現的恢復，進一步顯示與鐵死亡（ferroptosis）相關的 神經元損傷受到抑制。 綜合上述結果，標靶性 EV 所介導之 BDNF mRNA 遞送，可同時調控神經發炎、神經元可塑性以及氧化性損傷，支持其作為MDD無細胞治療策略（cell-free therapeutic strategy）的潛在應用價值

←Back to daily schedule



Plenary Keynote 2 - Speaker

Kuan-Pin Su (蘇冠賓), M.D., Ph.D.

Director, Mind-Body Interface Research Center, China Medical University Hospital
Deputy Superintendent, An-Nan Hospital, China Medical University, Tainan
(Taiwan)

PK-2 | 16:20- 17:00 | Abstract → Next

"Mind-Body Interventions and Psychoneuroimmune Modulation in Depression in the Digital Era"

CURRICULUM VITAE

Prof Kuan-Pin Su focuses on Translational Brain Research, connecting bedside to bench with novel interdisciplinary approaches of AI technology, neuroimaging, genomics, randomized- controlled trials (RCTs), cellular and molecular biology. He integrates clinical significance (e.g. pharmaceuticals, nutraceuticals, phytochemicals, rTMS, photobiomodulation, and acupuncture) with investigations of basic sciences. His significant contribution on the role of omega-3 polyunsaturated fatty acids (PUFAs) is opening the excitement and innovation therapeutic strategies and psychoneuroimmunity insights of depression. Prof Su's groundbreaking research has not only revolutionized the treatment guidelines for depression but has also pioneered the exciting field of nutritional psychiatry. A founding committee member and current Vice President of the International Society for Nutritional Psychiatry Research (ISNPR), Professor Su now leads the society's official journal as the Editor-in-Chief of Nutritional Psychiatry. According to the h-index (ELSVIER since 2018), Professor Su is the top-ranked principal investigator in the world in the research field of "omega-3 PUFAs in depression."

Prof Su has been involved in a series of clinical and basic research on major neuropsychiatric disorders

and earned internationally renowned reputation. He is the most highly cited scholar in depression research in Taiwan. Prof Su has received prestigious research awards from professional societies, including Robert Kerwin International Award from the British Association for Psychopharmacology (BAP, 2008), NARSAD Young Investigator Award (USA, 2008), International Society for the Study of Fatty Acids and Lipids (ISSFAL) New Investigator Award (2010 & 2012), National Science and Technology Council (NSTC) Ta-You Wu Memorial Award (2011), Thomson Reuters Research Front Awards (2011), Pacific Rim Psychiatrist College Young Psychiatrists Award (2012), Psychopharmacology Award from the BAP, UK (2013), ISSFAL Early Career Award (2016), China Medical University Distinguishing Professor (2018 & 2023), Taiwan NSTC Outstanding Research Award (2019), Y. Z. Hsu Science and Technology Chair Professor Award (2023), Taiwan Clinical TMS Society Distinguished Achievement Award (2024), and Psychoneuroimmunity Research Society (PNIRS) Norman Cousins Award.

Prof Su has been at the forefront of integrating artificial intelligence (AI), large language models (LLMs), and machine learning (ML) into psychiatric practice, education, and research. His team has developed digital tools for online screening, early detection of mental health warning signals, affective pattern recognition, digital phenotyping, and AI-assisted psychotherapy. He is the first to propose the concept of the “Digital Pandemic,” published in Psychiatry and Clinical Neurosciences and AI, conceptualizing the rapid deterioration of adolescent mental health in the digital era as a phenomenon with transmission dynamics analogous to infectious diseases. Committed to public service, Prof Su focuses on improving adolescent mental health and advancing equitable, dignified care through his work with the Ming-Yi Foundation, a non-profit organization, while also shaping national policy, cross-sectorcollaboration, and implementation as an appointed member of thePresidential Office’s Health Taiwan Committee.

←Back to Speaker CV

Mind-Body Interventions and Psychoneuroimmune Modulation in Depression in the Digital Era

Kuan-Pin Su (蘇冠賓), M.D., Ph.D.

Chairman, 16th MBI International Symposium
Director, Mind-Body Interface Research Center (MBI-Lab & Care), China Medical University
Professor of College of Medicine, China Medical University Deputy Superintendent of An-Nan Hospital, China Medical University

Abstract

Modern lifestyle factors have created an evolutionary mismatch that contributes to the rising global burden of depression, as biological systems optimized for ancestral survival are increasingly disrupted by chronic stress, physical inactivity, pro-inflammatory diets, circadian misalignment, and pervasive digital exposure. Chronic, low-grade systemic inflammation is a key mechanistic pathway linking these modern environmental pressures to depression and related disorders. Drawing on nearly 30 years of translational research, Prof. Kuan-Pin Su demonstrates how mind-body interventions can modulate neuroimmune and immunometabolic pathways relevant to depression in the digital era. Integrating evidence from molecular biology, placebo-controlled randomized trials, neuroimaging, and AI-enabled monitoring, this presentation highlights exercise, omega-3 fatty acids, melatonergic agents, acupuncture, and other lifestyle-based strategies as biologically informed interventions. Particular emphasis is placed on eicosapentaenoic acid (EPA) as a precision treatment for inflammation-related depression, as well as on the roles of the exercise–gut–brain axis, circadian regulation, and traditional interventions converging on shared anti-inflammatory mechanisms. Within this framework, the emerging “digital pandemic” is further considered a novel amplifier of evolutionary mismatch, especially in younger populations. Collectively, these findings support an integrative model of patient- centered precision psychiatry combining anti-inflammatory lifestyle medicine, digital therapeutics, and wearable technologies.

←Back to day 2 schedule



Plenary Keynote 3 - Speaker

Jane Pei-Chen Chang, M.D., M.Sc., Ph.D.

Director, Child Psychiatry Division, Department of Psychiatry, China
Medical University (CMU) Hospital, Taichung, Taiwan
President, Taiwanese Society for Nutritional Psychiatry Research(TSNPR)
(Taiwan)

PK- 3 | 08:30 - 09:10 | Abstract → Next

"From COVID-19 to the Digital Pandemic: Are We Raising a Generation Under Continuous Stress?"

CURRICULUM VITAE

Jane Pei-Chen Chang, M.D., M.Sc., Ph.D., is Director of the Child Psychiatry Division, Department of Psychiatry, China Medical University (CMU) Hospital, Taichung, Taiwan, and President of the Taiwanese Society for Nutritional Psychiatry Research (TSNPR). Her honors include the WFSBP Young Researcher's Award (2019), ASCAPAP Future of Asian Child and Adolescent Psychiatry Award (2019), Dr. Paul Janssen Research Awards (2015, 2020, 2024), Ta-You Wu Memorial Award (2021), ISSFAL New Investigator Awards (2014, 2018), and ISSFAL Early Career Award (2025). She has received research funding for clinical trials investigating omega-3 fatty acids, ADHD, depression, and autism spectrum disorder, as well as the Funds for Women Graduates Scholarship and NIHR/BRC Early Career Bridging and Career Development Grant in the UK (2018/2019). Dr. Chang has published 94 SCI-indexed papers. Her recent selected publications (2024–2026) address nutritional psychiatry implementation science, mother-child interactions in siblings of autistic children, omega-3 fatty acids in depression, obesity, COPD, COVID-19, and blood-brain barrier integrity.

←Back to Speaker CV

From COVID-19 to the Digital Pandemic: Are We Raising a Generation Under Continuous Stress?

Jane Pei-Chen Chang,
M.D., M.Sc., Ph.D.
Director, Child Psychiatry Division, Department of Psychiatry, China Medical University (CMU) Hospital, Taichung, Taiwan
President, Taiwanese Society for Nutritional Psychiatry Research(TSNPR)

Abstract

Young people today are growing up in a rapidly changing stress landscape. Adverse childhood experiences (ACEs) may establish early vulnerability to mental health difficulties, while the COVID-19 pandemic introduced a widespread layer of isolation, uncertainty, disrupted education, and family and social stress. As the pandemic receded, however, another transformation accelerated: the emergence of a Digital Pandemic. The Digital Pandemic refers to an evolving developmental environment in which adolescents and young adults increasingly experience relationships, information, entertainment, and emotional support through digital platforms. Social media algorithms, constant connectivity, misinformation, and increasingly sophisticated artificial intelligence may reshape how young people perceive themselves, relate to others, and respond to stress. At the same time, digital technologies can also provide connection, information, and new opportunities for support. This talk will discuss about how ACEs, COVID-19-related stress, and the emerging Digital Pandemic may influence mental health across development. Drawing on findings from adolescents and young adults, including evidence from Taiwan, the talk will explore whether we are moving from a period of acute pandemic stress toward an environment of continuous digital stress, and what this may mean for prevention, resilience, and the mental health of the next generation.

← Back to Index

S21. Gut–Brain Axis & Psychobiotics in Psychiatry

Chair: Jane Pei-Chen Chang (Taiwan)

Director, Child Psychiatry Division, Department of Psychiatry, China Medical University Hospital

Time: 09:10 - 10:30 (80 min)

Time	Speakers	Title
S 21- 1	Quang Le Do (Taiwan)	Omega-3 Reshape Gut Microbiota in Depression and ADHD
S 21- 2	Enchia Tsai (Taiwan)	Clinical Translation and Therapeutic Impact of Psychobiotic L. plantarum PS128 on Neurodevelopmental Health
S 21- 3	Wei-Hsiang Lin (Taiwan)	Translational Evidence of the Gerobiotic Lacticaseibacillus paracasei PS23 in Subjective Cognitive Decline: From Mechanistic Models to Human Outcomes
S 21- 4	Liang-Jen Wang (Taiwan)	Interaction of Candida albicans With Bacteria and Pathogenesis Mechanism in Attention-Deficit/Hyperactivity Disorder
10:30 - 10:50	Coffee Break @Corridor of Conference Hall	

←Back to Session 21 lineup



SESSION CHAIR

Jane Pei-Chen Chang (張倍禎), M.D., M.Sc., Ph.D.
Director, Child Psychiatry Division, Department of Psychiatry, China, Medical University (CMU) Hospital, Taichung, Taiwan, President, Taiwanese Society for Nutritional Psychiatry Research(TSNPR) **(Taiwan)**

S 21

09:10 - 10:30

S21: Psychobiotics and the Gut–Brain Axis in Neuropsychiatric Disorders: From Mechanisms to Precision Interventions

“ Summary of Session”

The Gut-Brain Axis and Psychobiotics session focuses on the advances in gut-brain axis research and the application of psychobiotics to mental health. Dr. Do Le Quang will present pilot data on the impact of Omega-3 fatty acids (eicosapentaenoic acid (EPA) and Docosahexaenoic Acid (DHA)) on the gut microbiota in patients with depression and attention deficit hyperactivity disorder (ADHD), showing that EPA tends to regulate differently compared to DHA. Ms. En- Chia Tsai will share a strategic perspective on the development of psychobiotics in biotechnology, highlighting the potential for application from research to clinical practice. Dr. Wei-Hsiang Lin will present evidence of a shift in the use of the Gerobiotic Lacticaseibacillus paracasei PS23, demonstrating its effectiveness in improving cognition and reducing neurodegenerative biomarkers in individuals with subjective cognitive impairment, while also confirming the mechanism in the C. elegans and 5xFAD mouse models. Prof. Liang-Jen Wang will report on research into bacteriofungal interactions in ADHD, particularly the role of Candida albicans in gut barrier dysfunction and its association with clinical symptoms. The session emphasizes that psychobiotics and complex microbiome interactions open up new prospects in the prevention and treatment of neuropsychiatric disorders. These preliminary results affirm the potential of the gut-brain axis as a platform for future precision medicine and lifestyle interventions.

←Back to Session 21 lineup



SPEAKER

Quang Le Do, M.D.

Ph.D. student, Graduate Institute of Nutrition, China Medical University

(Taiwan)

S 21-1 Abstract → Next

"Omega-3 Reshape Gut Microbiota in Depression and ADHD"

CURRICULUM VITAE

Quang Le Do, M.D., is a Ph.D. student at the Graduate Institute of Nutrition, China Medical University, Taichung, Taiwan, since February 2024. He has also been affiliated with the Mind-Body Interface Research Centre at China Medical University Hospital since April 2024. His research focuses on nutritional psychiatry, the gut-brain axis, gut microbiota, short-chain fatty acids, depression, and sleep disorders. He received the Best Poster Award at the International Society for Nutritional Psychiatry Research Conference 2025 (ISNPR 2025) for his study on the association of underweight with gait and predicted fall risk in patients with chronic obstructive pulmonary disease (COPD). His selected publications include research on circulating short-chain fatty acid profiles as potential biomarkers of gut-brain axis dysfunction in major depression, circadian rhythms and gut microbiota dysbiosis in insomnia, and the clinical and psychoneuroimmune implications of eicosapentaenoic acid in treatment-resistant depression.

←Back to Speaker CV

Omega-3 Reshape Gut Microbiota in Depression and ADHD

Quang Le Do, M.D.

Ph.D. student, Graduate Institute of Nutrition, China Medical University

Abstract

Background:

The gut microbiome and its metabolites play a crucial role in maintaining intestinal health, regulating immune responses, and influencing the central nervous system via the gut-brain axis. Recent studies have highlighted the potential impact of gut microbiota on mental health, including Major Depressive Disorder (MDD) and Attention Deficit Hyperactivity Disorder (ADHD). Changes in gut composition have been reported in patients with those disorders. Omega-3 polyunsaturated fatty acids, particularly eicosapentaenoic acid (EPA), may modulate gut microbiota, offering a novel mechanistic link of the gut-brain axis.

Methods:

The first pilot trial is a 12-week randomized, placebo-controlled trial in which 24 ADHD patients received daily supplementation with either 1 g of EPA (n=7), 1 g of docosahexaenoic acid (DHA) (n=8), or bean oil as a placebo (n=9). The second pilot trial is a 12-week randomized, placebo-controlled trial in 19 MDD patients with 3g n-3 PUFAs (EPA) in the intervention group (n=8) and bean oil in the placebo group (n=11). Stool samples were collected at baseline and post- intervention for analysis of the gut microbiome in both trials.

Results:

In patients with MDD and ADHD, an increasing trend of the Bacteroides phylum and a decreasing trend of Firmicutes have been shown, but not significantly. EPA supplementation showed a significant decrease in the Bacteroides phylum in ADHD patients compared to controls, but DHA showed the opposite effect. In MDD patients, the effects of EPA on Bacteroides and Firmicutes were unclear. EPA, however, tended to increase Actinobacteria, particularly Bifidobacterium, a key beneficial taxon.

Conclusion: Preliminary data indicate that EPA and DHA may modulate gut microbiota in MDD and ADHD patients in different ways. Additionally, beneficial effects were shown in the EPA group. However, the effects are not clear due to the small sample size. Further studies with larger sample sizes are warranted to clarify the clinical implications of these differential effects.

←Back to Session 21 lineup



SPEAKER

Enchia Tsai (蔡恩加), Ms.c.

Co-Founder and Executive Director, Head of Strategy and Innovation, Bened Biomedical Co., Ltd.

(Taiwan)

S 21-2

Abstract → Next

"Clinical Translation and Therapeutic Impact of Psychobiotic L. plantarum PS128 on Neurodevelopmental Health"

CURRICULUM VITAE

Enchia Tsai (蔡恩加), M.Sc., is the Co-Founder and Executive Director, as well as Head of Strategy and Innovation at Bened Biomedical Co., Ltd. She has been recognized for her professional achievements and contributions to the biotechnology sector. Her honors include the Study UK Alumni Awards – Global Winner, making her the first Taiwanese recipient in recognition of her outstanding professional achievements and contributions to UK–Taiwan academic and business exchange. In 2017, she also received the Best Female Entrepreneur (Startups) Award in Taiwan for her leadership and innovation in the biotechnology industry.

←Back to Speaker CV

Clinical Translation and Therapeutic Impact of Psychobiotic L. plantarum PS128 on Neurodevelopmental Health

Enchia Tsai (蔡恩加), Ms.c.

Co-Founder and Executive Director, Head of Strategy and Innovation, Bened Biomedical Co., Ltd.

Abstract

The bidirectional communication within the gut-brain axis (GBA) has opened a new frontier in neurology and psychiatry, introducing targeted microbial interventions as significant adjunctive therapeutic options. Among these, the psychobiotic Lactobacillus plantarum PS128 has demonstrated a unique capacity to modulate dopaminergic and serotonergic pathways, offering a non- pharmacological approach to supporting neurological health. This presentation reviews the clinical evidence of PS128, focusing on its role as a therapeutic adjuvant for neurodevelopmental conditions such as Autism Spectrum Disorder (ASD) and Attention Deficit Hyperactivity Disorder (ADHD). Data from multiple clinical trials suggest that PS128 can effectively alleviate core symptoms—including social anxiety, emotional dysregulation, and hyperactivity—while offering a high safety profile that complements traditional pharmacological treatments. Beyond the clinical data, we discuss the strategic innovation required to translate these findings into a validated therapeutic choice. By bridging the gap between microbial research and clinical practice, PS128 serves as a prime example of how strain-specific psychobiotics are evolving from general supplements into precision neuro-healthcare tools. This “Lab to Life” transition provides clinicians and families with a evidence-based, integrative option to enhance cognitive well-being and long-term developmental outcomes.

←Back to Session 21 lineup



SPEAKER

Wei-Hsiang Lin (林暐翔), Ph.D.

Chief Scientist, Bened Biomedical Co., Ltd.

(Taiwan)

S 21-3

Abstract → Next

"Translational Evidence of the Gerobiotic Lacticaseibacillus paracasei PS23 in Subjective Cognitive Decline: From Mechanistic Models to Human Outcomes"

CURRICULUM VITAE

Wei-Hsiang Lin (林暐翔), Ph.D., is the Chief Scientist at Bened Biomedical Co., Ltd., Taiwan, a position he has held since September 2024. Previously, he served as an Experimental Officer in the Genome Editing Unit at the University of Manchester, UK (March 2022–August 2024), and as a Research Fellow at the University of Manchester (November 2006–February 2022). His research has been supported by several prestigious UK research grants, including funding from the Leverhulme Trust (2018–2022) and the Biotechnology and Biological Sciences Research Council (BBSRC) (2007–2017). His research focuses on neuroscience, genetics, neuronal homeostasis, seizure mechanisms, and molecular biology. He has authored a total of 25 SCI-indexed publications. His selected publications include studies published in The Lancet Neurology, Nature, Brain, Disease Models & Mechanisms, and the European Journal of Neuroscience, covering genetic neuropathy, seizure suppression, neuronal regulation, and micronutrient sensing.

←Back to Speaker CV

Translational Evidence of the Gerobiotic Lacticaseibacillus paracasei PS23 in Subjective Cognitive Decline: From Mechanistic Models to Human Outcomes

Wei-Hsiang Lin (林暐翔), Ph.D.

Chief Scientist, Bened Biomedical Co., Ltd.

Abstract

Background:

Subjective cognitive decline (SCD) is increasingly recognized as an early stage of Alzheimer's disease, yet effective interventions remain limited. Microbiota-based approaches targeting the gut-brain axis have emerged as potential strategies to modulate neurodegenerative processes. Lacticaseibacillus paracasei PS23 (PS23), a candidate gerobiotic, has previously demonstrated cognitive benefits in aging-related models. Methods: Individuals with SCD participated in a 12-week intervention study with PS23 supplementation. Cognitive assessments included the Subjective Cognitive Decline Scale (SCDS), Trail Making Test-B (TMT-B), Stroop Color and Word Test (SCWT), and CANTAB cognitive battery. Blood biomarkers associated with neurodegeneration, including amyloid-β40 (Aβ40), amyloid-β42 (Aβ42), glial fibrillary acidic protein (GFAP), and neurofilament light chain (NfL), were also evaluated. Complementary mechanistic studies were conducted using Caenorhabditis elegans GMC101 and 5xFAD transgenic mouse models.

Results:

PS23 significantly improved total SCDS scores and executive, memory, and language domains. Improvements were also observed in processing speed, executive function, memory performance, and Stroop test performance. Biomarker analyses showed significant reductions in Aβ40, Aβ42, GFAP, and NfL, while the Aβ42/40 ratio showed an increasing trend. In preclinical models, PS23 delayed amyloid-β-induced paralysis in GMC101 worms and improved cognition, reduced amyloid burden, and suppressed neuroinflammation in 5xFAD mice.

Conclusion:

PS23 demonstrated consistent cognitive and biomarker benefits across clinical and preclinical models, supporting its potential as a gerobiotic intervention for early cognitive decline.

←Back to Session 21 lineup



SPEAKER

Liang-Jen Wang (王亮人), M.D., M.P.H., Ph.D.

Director, Department of Psychiatry, Kaohsiung Chang Gung Memorial Hospital

(Taiwan)

S 21-4

Abstract → Next

"Interaction of Candida albicans With Bacteria and Pathogenesis Mechanism in Attention-Deficit/Hyperactivity Disorder"

CURRICULUM VITAE

Liang-Jen Wang (王亮人), M.D., M.P.H., Ph.D., is the Director of the Department of Psychiatry at Kaohsiung Chang Gung Memorial Hospital, Taiwan, and a Professor at the School of Medicine, Chang Gung University. He has received numerous research and academic awards, including the Best Research Award at the 12th Congress of the Asian Society for Child and Adolescent Psychiatry and Allied Professions (ASCAPAP 2026), the Excellence Award for Academic Publications from Kaohsiung Chang Gung Memorial Hospital (2025), the Outstanding Paper Award from the Taiwanese Society of Psychiatry (2025), and the Paper Award from the Taiwan Society of Addiction Science (2024). He also received the Chang Wen-Ho Research Paper Award (2024) and the Dr. Paul Janssen Outstanding Research Paper Award (2024). He is a member of several professional organizations, including the Psychiatric Society of Taiwan, the Child and Adolescent Psychiatric Society of Taiwan, the Taiwanese Society of Biological Psychiatry and Neuropsychopharmacology, the Society of Biological Psychiatry (USA), and the Taiwanese Society of Addiction Science.

←Back to Speaker CV

Interaction of Candida albicans With Bacteria and Pathogenesis Mechanism in Attention Deficit/Hyperactivity Disorder

Liang-Jen Wang (王亮人), M.D., M.P.H., Ph.D.

Director, Department of Psychiatry, Kaohsiung Chang Gung Memorial Hospital

Abstract

Objective:

Attention-deficit/hyperactivity disorder (ADHD) is one of the most common neurodevelopmental disorders in children. Although previous studies have linked ADHD with gut bacteria and fungi, the potential interactions between these microbial communities remain unclear. Method: We applied full-length 16S rRNA and internal transcribed spacer (ITS) sequencing to stool samples from individuals with ADHD and healthy controls (HCs) to investigate bacteria–fungi interactions and their relevance to ADHD. Full-length 16S rRNA sequencing provided higher resolution for bacterial identification than conventional V3-V4 sequencing. Microbial profiles based on both bacterial and fungal abundances clearly distinguished ADHD from HC samples.

Results:

Network analyses revealed significant cross-kingdom interactions, such as a negative association between Candida albicans and the bacterial genus Bacteroides, and positive associations between Auricularia nigricans and several bacterial species, including Alistipes senegalensis, Bacteroides faecis, and Eubacterium ventriosum. These relationships suggest that metabolite-driven interactions may shape a symbiotic gut microenvironment. Among fungal species, C. albicans showed the greatest difference between ADHD and HC groups. Its abundance correlated strongly with ADHD clinical features,

including full-scale intelligence quotient, perceptual reasoning index, and the severity of inattention and hyperactivity/impulsivity symptoms. In vitro assays further demonstrated that C. albicans compromised gut epithelial integrity by reducing cell-cell junctions, inhibiting proliferation, and promoting apoptosis.

Conclusion:

These findings highlight the importance of bacteria–fungi interactions in ADHD and suggest that C. albicans may contribute to both microbial imbalance and gut barrier dysfunction associated with the disorder.

Keywords: ADHD, bacteria, fungus, gut-brain axis, microbiome, gut leakage

← Back to Index

S22. Nutritional Psychiatry: Clinical Insights from Japan

Co-Chair: Ryohei Takeno (Japan); Shizuka Ota, M.D.

Time: 10:50 - 12:00 (70 min)

Time	Speakers	Title
S 22- 1	Tomoyuki Okudaira, M.D (Japan)	Development of Nutritional Psychiatry in Japan: From Clinical Experience to Academic Society Formation
S 22- 2	Yohei Kita, M.D., Ph.D. (Japan)	Suspected Thiamine Deficiency in Malnourished Psychiatric Inpatients: Early Recognition and Empirical Nutritional Intervention
S 22- 3	Hiroyuki Konno, M.D., Ph.D. (Japan)	Cognitive Improvement in an 83-Year-Old Male With Mild Cognitive Impairment After Multimodal Nutritional and Metabolic Intervention Including Exogenous Ketone Supplementation: A Case Report
12:00 - 13:00	Lunch @B202, B203	

←Back to Session 22 line up

SESSION CHAIR

Co-Chair: Ryohei Takeno
Deputy Director, Depertment of Psychiatry, Saitama Sekishinkai Hospital **(Japan)**

Co-Chair: Shizuka Ota, M.D.
Director and Chairperson, Watanabe Mental Clinic, Medical Corporation, Chishinkai
Secretariat, Japanese Society of Nutritional Psychiatry **(Japan)**

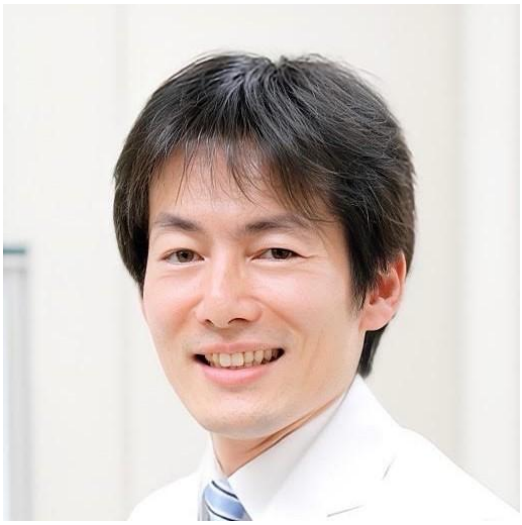
S 22 **10:50 - 12:00**

S22: Japanese Clinical Perspectives on Nutritional Psychiatry at the Mind-Body Interface: Micronutrients, Metabolism, and Cognitive Health

" Summary of Session"

This symposium presents Japanese clinical perspectives on integrating nutritional assessment into psychiatric diagnosis and standard treatment, focusing on micronutrients, metabolism, and cognition. The first outlines the Okudaira-led development of nutritional psychiatry in Japan. In 2014, Okudaira coined the Japanese term Eiyo Seishin Igaku and led nationwide education for healthcare professionals and the public. The Japan Nutritional Psychiatry Research Group, founded in 2016, held 25 multidisciplinary meetings through 2025. In October 2025, it launched the peer-reviewed Journal of the Japanese Society of Nutritional Psychiatry and declared its transition to a society. On April 1, 2026, a non-profit two-corporation structure was established: the Japanese Society of Nutritional Psychiatry (JSNP), for academic activities and certification, and the Japanese Association of Nutritional Psychiatry (JANP), for education. Its certification system comprises specialist and public-oriented qualifications and accredited medical and counseling facilities. As of June 2026, JSNP had over 1,000 members (over one-third physicians), 150 delegates (132 psychiatrists), 82 facilities, and 282 certified individuals holding 453 certifications, illustrating an organizational and educational foundation in Asia. The second reports a retrospective study of undernourished psychiatric inpatients with suspected thiamine deficiency, in whom empirical nutritional intervention was associated with marked MMSE improvement and low delirium incidence, underscoring early recognition. The third describes an 83-year-old man with mild cognitive impairment whose cognition improved after multimodal nutritional and metabolic intervention including exogenous ketones. The session highlights how nutritional and metabolic interventions can complement psychiatric and cognitive care, emphasizing evidence-based practice, cautious causal interpretation, multidisciplinary collaboration, and future research in Asia.

←Back to Session 22 line up



SPEAKER

Tomoyuki Okudaira, M.D.

Founder and President, The Japanese Society of Nutritional Psychiatry
(Japan)

S 22-1

Abstract → Next

"Development of Nutritional Psychiatry in Japan: From Clinical Experience to Academic Society Formation"

CURRICULUM VITAE

Tomoyuki Okudaira, M.D., is the Founder and President of the Japanese Society of Nutritional Psychiatry (JSNP) and the Japanese Association of Nutritional Psychiatry (JANP), and Vice Director of Yamaguchi Hospital, Japan. Since 2014, he has initiated and led nationwide educational activities in nutritional psychiatry across Japan, including invited lectures and seminars for healthcare professionals and the general public. From 2016 to 2025, he served as Founder and President of the Japan Nutritional Psychiatry Research Group. Since 2026, he has led the JSNP and JANP. He is an Editorial Committee Member and Reviewer for Nutritional Psychiatry, the official journal of the International Society for Nutritional Psychiatry Research (ISNPR), and Editor-in-Chief of Nutritional Psychiatry, the official journal of JSNP. He also serves as a Delegate of the Japanese Society of Psychiatry and Neurology. His selected publications focus on nutritional psychiatry, including the roles of zinc, iron, kynurenine, and magnesium in mental health, as well as nutritional approaches to clinical psychiatric care. His additional contributions include a 12-part review series on nutritional psychiatry, a case report, and five single-authored books widely used for public health education in Japan, with a cumulative circulation of approximately 140,000 copies.

←Back to Speaker CV

Development of Nutritional Psychiatry in Japan: From Clinical Experience to Academic Society Formation

Tomoyuki Okudaira, M.D.

Founder and President, The Japanese Society of Nutritional Psychiatry

Abstract

Background:

Nutritional psychiatry is a clinical field that scientifically positions nutritional assessment within psychiatric diagnosis and standard treatment. In Japan, a dedicated organizational and academic foundation for this field was long absent. Method: This presentation describes the development from the Japan Nutritional Psychiatry Research Group—proposed by Okudaira in 2014 and founded in 2016—to the incorporated Japanese Society of Nutritional Psychiatry (JSNP), and its certification system, using organizational and educational indicators.

Result:

In 2014, Okudaira coined the Japanese term Eiyo Seishin Igaku and led nationwide educational activities for healthcare professionals and the public. The research group held 25 scientific meetings through 2025, examining clinically relevant topics across disciplines. On October 12, 2025, at its 10th-anniversary meeting, it launched the peer-reviewed Journal of the Japanese Society of Nutritional Psychiatry and declared its transition to a society. On April 1, 2026, a non-profit two-corporation structure was established: JSNP, responsible for academic activities and certification, and the Japanese Association of Nutritional Psychiatry (JANP), responsible for education. The certification system comprises specialist qualifications for physicians, dentists, and licensed healthcare and nutrition professionals; public-oriented qualifications; and accredited medical and counseling facilities, renewed triennially. As of June 2026, membership exceeded 1,000 (over one-third physicians), with 150 delegates (132 psychiatrists), 82 accredited facilities, and 282 certified individuals holding 453 certifications.

Conclusion:

Under Okudaira's leadership, Japanese nutritional psychiatry evolved from individual clinical efforts into an organized, diagnosis-based academic field. Future priorities are translating this quantitative growth into clinical quality assurance and internationally disseminating evidence from Japanese populations.

←Back to Session 22 line up



SPEAKER

Yohei Kita, M.D., Ph.D.

Vice Director, Department of Psychiatry, Shomidori Hospital
(Japan)

S 22-2

Abstract → Next

"Suspected Thiamine Deficiency in Malnourished Psychiatric Inpatients: Early Recognition and Empirical Nutritional Intervention"

CURRICULUM VITAE

Yohei Kita, M.D., Ph.D., is Vice Director of the Department of Psychiatry at Shomidori Hospital, Japan, a position he has held since 2015. Previously, he worked as a psychiatrist at Juntendo University Hospital (2008–2014) and completed his psychiatric residency training at Dokkyo Medical University Hospital (2005–2007). He also serves as Director and Editorial Committee Member of the Japanese Society of Nutritional Psychiatry. He is a Designated Mental Health Physician under Japan's Mental Health and Welfare Act and a member of the Japanese Society of Psychiatry and Neurology. His research and publications include studies on serum amyloid β protein in young and elderly patients with depression and clinical and diagnostic considerations of schizotypal personality disorder, particularly its relationship with schizophrenia and borderline personality disorder.

←Back to Speaker CV

Suspected Thiamine Deficiency in Malnourished Psychiatric Inpatients: Early Recognition and Empirical Nutritional Intervention

Yohei Kita, M.D., Ph.D.
Vice Director, Department of Psychiatry, Shomidori Hospital

Abstract

Background:

Thiamine deficiency is underrecognized in malnourished psychiatric inpatients, with clinical profile and short-term outcomes poorly characterized, including in non-alcoholic populations (Mates et al., 2021). This study evaluated clinical features and outcomes in such patients receiving empirical nutritional intervention. Method: This single-center retrospective observational study included 28 psychiatric inpatients (ICD-10 F2/F3) admitted in 2025 with clinically suspected thiamine deficiency, identified by neurological examination (gait disturbance, tendon reflex changes), nutritional assessment, skin findings, and dietary history. Dementia was excluded by a neurologist; one patient with identified alcohol use disorder was excluded. Empirical intervention comprised ENEFLUID® peripheral parenteral multi-nutrient infusion (up to 4 bags/day) plus oral thiamine (≥200 mg/day), transitioned to oral preparations as oral intake recovered. MMSE was assessed at admission and one month. Ethics committee approval was obtained; consent was waived.

Result:

Twenty-seven patients (15 men, 12 women; mean age 54.5 years) were analyzed. Gait disturbance with reduced/absent tendon reflexes was present in 5; short-term memory impairment or disorientation in 15. Delirium or disturbance of consciousness occurred in 2 patients (7.4%). Among 15 with cognitive symptoms, MMSE improved from median 14.0 (IQR 12.5–16.0) to 29.0 (IQR 28.0–29.5) at one month (median +15.0; Wilcoxon signed-rank test, $p = 0.001$).

Conclusion:

Systematic neurological and nutritional assessment may facilitate early identification of suspected thiamine deficiency in psychiatric inpatients. Parenteral multi-nutrient support and oral thiamine appeared to facilitate recovery, though concurrent psychiatric treatment, environmental stabilization, and dietary improvement preclude attribution to specific interventions. As an uncontrolled study without biochemical confirmation, findings are hypothesis- generating; prospective controlled studies are warranted.

←Back to Session 22 line up



SPEAKER

Hiroyuki Konno, M.D., Ph.D.

Honorary Director, Brain Care Clinic
(Japan)

S 22-3

Abstract → Next

"Cognitive Improvement in an 83-Year-Old Male With Mild Cognitive Impairment After Multimodal Nutritional and Metabolic Intervention Including Exogenous Ketone Supplementation: A Case Report"

CURRICULUM VITAE

Hiroyuki Konno, M.D., Ph.D., is the Honorary Director of Brain Care Clinic and former Chief Director of the Japan Brain Care and Dementia Prevention Research Center, Japan. He served as Honorary Director of Brain Care Clinic from 2019 to 2026 and Chief Director of the Japan Brain Care and Dementia Prevention Research Center from 2018 to 2026. Previously, he worked as a psychiatrist at Kunpukai Yamada Hospital from 2007 to 2016. He is a Delegate of the Japanese Society of Nutritional Psychiatry and holds specialist qualifications in psychiatry from the Japanese Society of Psychiatry and Neurology and anti-aging medicine from the Japanese Society of Anti-Aging Medicine. His selected publications include research on the effects of GABA and L-theanine supplementation on sleep quality, nutritional supplementation and cognitive function in middle-aged and older adults, and the effects of melinjo seed extract on serum uric acid levels.

←Back to Speaker CV

Cognitive Improvement in an 83-Year-Old Male With Mild Cognitive Impairment After Multimodal Nutritional and Metabolic Intervention Including Exogenous Ketone Supplementation: A Case Report

Hiroyuki Konno, M.D., Ph.D.

Honorary Director, Brain Care Clinic

Abstract

Background:

Alzheimer's disease and mild cognitive impairment (MCI) are multifactorial conditions where reversible factors—including nutritional deficiencies, glucose dysregulation, digestive dysfunction, inflammation, and environmental exposures—may contribute to cognitive decline. Exogenous ketone supplementation has emerged as a potential adjunctive strategy, though clinical evidence remains limited. Method: An 83-year-old male with MCI underwent a multimodal intervention involving nutritional and metabolic assessment. This included a low-GI diet, 12- hour intermittent fasting, restriction of carbohydrates, gluten, and casein, protein intake ≥1.5 g/kg/day, alcohol cessation, supplementation (vitamin B complex, C, D, and glutamine), aerobic exercise, sleep hygiene counselling, and reduction of environmental exposures. After six months, 4 g of D-BHB (exogenous ketone) was added, with cognitive assessment performed four months later.

Result:

At baseline, the patient had an MMSE score of 28 and a CNS Vital Signs Neurocognitive Index score of 85. Laboratory findings revealed hyperglycemia and positive serum Helicobacter pylori antibodies, suggestive of atrophic gastritis. Additionally, homocysteine was 12.6 μmol/L and 25-OH vitamin D was 21.9 ng/mL, suggesting nutritional insufficiency. After six months of multimodal intervention, laboratory findings partially improved, but cognitive scores showed no marked change. Four months after adding exogenous ketone supplementation, the Neurocognitive Index score improved to 98.

Conclusion:

This case suggests cognitive improvement may occur when adding exogenous ketone supplementation to multimodal nutritional and metabolic interventions in MCI. As this is a single case with concurrent interventions, causal effects cannot be determined. Further clinical studies are needed to clarify the role of exogenous ketones in cognitive health.

← Back to Index

S23. Rewiring the Inflamed Mind: Immuno-Psychiatry

Co-Chair: Piotr Gałecki, M.D., Ph.D. & Co-Chair: Marta Broniarczyk-Czarniak, M.D., Ph.D.

Time: 13:00 - 14:40 (100 min)

Time	Speakers	Title
S 23- 1	Piotr Gałecki, M.D., Ph.D. (Poland)	From Inflammatory Depression to Personalized Immunopsychiatry: Clinical Standards and Translational Pathways
S 23- 2	Marta Broniarczyk-Czarniak, M.D., Ph.D. (Poland)	Neuropeptides, Social Stress, and the Immune-Brain Interface in Severe Mental Disorders
S 23- 3	Gracja Ryznar, M.D. (Poland)	Cytokines and the Mind: Immunological Underpinnings of Depression and Anxiety Symptoms
S 23- 4	Katarzyna Bliźniewska-Kowalska, M.D., Ph.D. (Poland)	Breaking the Brain's Resilience: How Inflammatory Pathways Impair Neuroplasticity
S 23- 5	Małgorzata Gałecka, Ph.D. (Poland)	Healing Beyond the Mind: The Impact of Psychotherapy on the Immunological Profile
14:40 - 15:00	Coffee Break	@Corridor of Conference Hall

←Back to Session 23 line up

SESSION CHAIR

Co-Chair: Piotr Gałecki, M.D., Ph.D.
Professor and Head, Department of Adult Psychiatry, Medical University of Łódź
Co-Chair: Marta Broniarczyk-Czarniak, M.D., Ph.D.
Assistant Professor, Department of Adult Psychiatry, Medical University of Łódź **(Poland)**

S - 23 **13:00 - 14:40**

S23: Rewiring the Inflamed Mind: Neuroplasticity, Mood Disorders, and Psychotherapy Within the Immune-Brain Axis

" Summary of Session"

This thematic session presents a translational framework for immunopsychiatry, linking inflammatory pathways, neuroplasticity, psychotherapy, and clinical innovation in mood and stress-related disorders. The session begins with cytokine dysregulation in depression and anxiety, with particular attention to IL-17/Th17 signaling, anti-inflammatory imbalance, blood-brain barrier integrity, HPA-axis activation, and downstream effects on neurotransmission. It then examines how chronic low-grade inflammation disrupts neuroplasticity through BDNF suppression, kynurenine-pathway activation, glutamatergic excitotoxicity, and impaired hippocampal neurogenesis, while also discussing ketamine as a rapid- acting intervention with potential anti-inflammatory and synaptogenic effects. A third presentation focuses on how integrated psychiatric and psychotherapeutic treatment may influence anxiety, depressive symptoms, and broader mind-body outcomes. The session further expands the immune-brain discussion to neuropeptide systems implicated in social stress, resilience, and severe mental disorders. The concluding presentation synthesizes these perspectives into a personalized clinical model, emphasizing biomarkers, treatment-resistant depression, and the practical translation of immune-brain research into psychiatric standards of care.

←Back to Session 23 line up



SPEAKER

Piotr Gałecki, M.D., Ph.D.

Professor and Head, Department of Adult Psychiatry,
Medical University of Łódź
(Poland)

S 23-1 Abstract → Next

From Inflammatory Depression to Personalized Immunopsychiatry: Clinical Standards and Translational Pathways

CURRICULUM VITAE

Piotr Gałecki, M.D., Ph.D., is a Professor and Head of the Department of Adult Psychiatry at the Medical University of Łódź, Poland, and serves as the National Consultant in Psychiatry in Poland. He has been engaged in academic psychiatry and clinical leadership at the Medical University of Łódź since 2007. Since 2016, he has held national expert roles in developing psychiatric standards, treatment recommendations for treatment-resistant depression, and mental health reform. His research focuses on the inflammatory, neurodevelopmental, oxidative, and treatment-resistant models of depression. He holds an M.D. from the Medical University of Łódź and is a Professor of Medical Sciences and specialist in psychiatry and sexology. He has authored or co-authored more than 300 scientific publications and psychiatric textbooks. His selected PubMed-indexed publications address the inflammatory and neurodevelopmental theories of depression, the anti-inflammatory mechanisms of antidepressants, oxidative and nitrosative stress pathways in major depression, and Polish standards for racemic ketamine treatment in patients with depressive disorders.

←Back to Speaker CV

From Inflammatory Depression to Personalized Immunopsychiatry: Clinical Standards and Translational Pathways

Piotr Gałecki, M.D., Ph.D.

Professor and Head, Department of Adult Psychiatry, Medical University of Łódź

Abstract

Depression is increasingly understood as a heterogeneous disorder in which immune-inflammatory activation, oxidative and nitrosative stress, stress-system dysregulation, altered neuroplasticity, and adverse developmental trajectories interact with psychosocial vulnerability. This concluding presentation integrates the session's mechanistic and clinical themes into a practical model of personalized immunopsychiatry. It will review the inflammatory and neurodevelopmental theories of depression, the relevance of anti-inflammatory mechanisms of antidepressants, and the clinical importance of treatment-resistant depression. Particular attention will be paid to how biomarkers should be used in psychiatry: not as stand-alone diagnostic substitutes, but as tools that may enrich clinical formulation, identify biologically distinct subgroups, and support rational treatment selection. The presentation will also discuss the translational pathway from biological findings to clinical standards, including recommendations for treatment-resistant depression and the emerging role of racemic ketamine in depressive disorders. The conclusion proposes an integrative treatment algorithm in which pharmacotherapy, psychotherapy, neuromodulation, lifestyle interventions, and immune-brain biomarkers are considered within one coherent clinical framework. This approach supports the overall goal of the session: to move immunopsychiatry from laboratory evidence toward individualized, clinically meaningful care.

←Back to Session 23 line up



SPEAKER

Marta Broniarczyk-Czarniak, M.D., Ph.D.

Assistant Professor, Department of Adult Psychiatry,
Medical University of Łódź
(Poland)

S 23-2

Abstract → Next

"Neuropeptides, Social Stress, and the Immune-Brain Interface in Severe Mental Disorders"

CURRICULUM VITAE

Marta Broniarczyk-Czarniak, M.D., Ph.D., is an Assistant Professor in the Department of Adult Psychiatry at the Medical University of Łódź, Poland. She is a specialist psychiatrist and researcher with clinical and academic experience in adult psychiatry since 2014. Her research focuses on the biological mechanisms, cognitive aspects, and clinical features of severe mental disorders. Since 2022, she has been involved in molecular research investigating neuropeptides and their role in psychosis-related symptoms. Her selected publications include studies on the role of oxytocin (OXT), oxytocin receptor (OXTR), arginine vasopressin (AVP), and vasopressin receptor 1A (AVPR1a) gene expression in schizophrenia, the epidemiology of suicide in bipolar disorder, mental disorders coexisting with Hashimoto's disease, and severe depression associated with cognitive impairment.

←Back to Speaker CV

Neuropeptides, Social Stress, and the Immune-Brain Interface in Severe Mental Disorders

Marta Broniarczyk-Czarniak, M.D., Ph.D.

Assistant Professor, Department of Adult Psychiatry, Medical University of Łódź

Abstract

The immune-brain axis is not limited to classical cytokine signaling; it also involves neuropeptide systems that regulate attachment, social cognition, stress response, and resilience. This presentation focuses on oxytocin and vasopressin pathways as biologically plausible bridges between interpersonal stress, emotional regulation, and severe mental disorders. Evidence from molecular psychiatric studies indicates that altered expression of OXT, OXTR, AVP, and AVPR1a may be associated with schizophrenia course and with depressive symptoms in psychotic disorders. These findings are discussed in the broader context of HPA-axis dysregulation, inflammatory activation, social defeat, and impaired neuroplastic adaptation. The presentation will also address how immune-inflammatory and neuropeptide mechanisms may converge clinically: patients exposed to chronic stress often present with disturbed social functioning, reduced resilience, sleep disruption, anxiety, depressive symptoms, and cognitive impairment. Understanding these pathways may help clinicians move beyond symptom clusters toward a more dimensional model of severe mental disorders. The conclusion emphasizes that neuropeptide and immune markers should be interpreted as complementary components of a personalized psychiatric assessment rather than isolated diagnostic tests. Such an approach may support future biomarker-informed interventions targeting social stress, resilience, and recovery.

←Back to Session 23 line up



SPEAKER

Gracja Ryznar, M.D.

Psychiatry resident, Department of Adult Psychiatry,
Medical University of Łódź
(Poland)

S 23-3

Abstract → Next

"Cytokines and the Mind: Immunological Underpinnings of Depression and Anxiety Symptoms"

CURRICULUM VITAE

Gracja Ryznar, M.D., is a Psychiatry Resident in the Department of Adult Psychiatry at the Medical University of Łódź, Poland (2025–Present). Since 2024, she has been engaged in early-career research activities focusing on mood disorders, inflammation, and ketamine treatment, as well as clinical and academic collaboration within the Medical University of Łódź team. Her selected publications include a 2025 case report on the potentiation of antidepressant treatment with racemic ketamine in a patient with severe depression in the course of bipolar disorder and a history of psychotic mania, published in Pharmacotherapy in Psychiatry and Neurology (41(2):157–168); a 2024 study on the correlation between depression, physical exercise, and psoriasis, published in Quality in Sport (16); and a 2024 publication on the role of physical exercise in diseases treated with colchicine, also published in Quality in Sport.

←Back to Speaker CV

Cytokines and the Mind: Immunological Underpinnings of Depression and Anxiety Symptoms

Gracja Ryznar, M.D.

Psychiatry resident, Department of Adult Psychiatry, Medical University of Łódź

Abstract

Growing evidence highlights a critical link between the immune system and central nervous system pathologies, specifically mood and anxiety disorders. This presentation explores the immunological underpinnings of depressive and anxiety symptoms, focusing on the systemic cytokine profile characterized by a profound imbalance between pro- and anti-inflammatory mediators. While diminished anti- inflammatory signaling fails to counter chronic low-grade inflammation, the hyperactivation of pro-inflammatory pathways—with a prominent role of the Th17 cell axis—drives psychiatric pathology. Interleukin-17 (IL-17), the signature cytokine of this pathway, actively promotes neuroinflammation by directly compromising blood-brain barrier (BBB) integrity through the disruption of endothelial tight junctions. This peripheral-to-central cascade alters neurotransmitter metabolism, impairs neuroplasticity via the kynurenine pathway, and hyperactivates the hypothalamic-pituitary-adrenal (HPA) axis. Clinical and preclinical data demonstrate that this neuroimmune dysregulation triggers microglial activation, manifesting as core depressive and anxious behaviors. Understanding this specific cytokine fingerprint is essential for identifying novel biomarkers and developing targeted, personalized immunotherapies for treatment-resistant psychiatric conditions.

←Back to Session 23 line up



SPEAKER

Katarzyna Bliźniewska-Kowalska, M.D., Ph.D.

Assistant Professor, Department of Adult Psychiatry,
Medical University of Łódź **(Poland)**

S 23-4

Abstract → Next

" Breaking the Brain's Resilience: How Inflammatory Pathways Impair Neuroplasticity"

CURRICULUM VITAE

Katarzyna Bliźniewska-Kowalska, M.D., Ph.D., is an Assistant Professor in the Department of Adult Psychiatry at the Medical University of Łódź, Poland. She is a specialist psychiatrist and researcher focusing on mood disorders and treatment-resistant depression. Since 2016, she has been involved in academic and clinical work at the Medical University of Łódź. Since 2021, she has contributed to the development of national recommendations and standards for the treatment of treatment-resistant depression and ketamine therapy. Since 2019, her research has focused on the molecular, immune-inflammatory, and oxidative mechanisms underlying depressive disorders. She has authored approximately 65 scientific publications, including papers indexed in PubMed and Science Citation Index (SCI), covering the molecular mechanisms of depression, gene expression, inflammatory pathways, and treatment-resistant depression.

←Back to Speaker CV

Breaking the Brain's Resilience: How Inflammatory Pathways Impair Neuroplasticity

Katarzyna Bliźniewska-Kowalska, M.D., Ph.D.

Assistant Professor, Department of Adult Psychiatry, Medical University of Łódź

Abstract

Brain resilience—the capacity to adapt to stress and resist neuropsychiatric pathologies—relies fundamentally on robust neuroplastic mechanisms. This presentation examines how chronic, low-grade inflammatory pathways disrupt this protective framework, ultimately leading to cognitive and emotional dysfunction. Central to this impairment is the detrimental effect of pro-inflammatory cytokines on neurotrophic support, particularly through the suppression of Brain-Derived Neurotrophic Factor (BDNF). Furthermore, neuroinflammation hyperactivates the kynurenine pathway, shifting tryptophan metabolism toward neurotoxic metabolites that induce glutamate excitotoxicity and synaptic pruning. At the cellular level, sustained inflammatory signalling compromises adult hippocampal neurogenesis and alters dendritic spine morphology. Crucially, this presentation highlights the therapeutic potential of rapid-acting interventions like ketamine, which has been shown to bypass inflammatory blocks, acutely suppress neuroinflammation, and rapidly restore synaptic density and BDNF expression. By unravelling these molecular cascades, we underscore how inflammatory pathways erode neural resilience and emphasize the critical need for novel, fast-acting strategies to restore neuroplasticity in stress-related and treatment-resistant disorders.

←Back to Session 23 line up



SPEAKER

Małgorzata Gałecka, Ph.D.
Assistant Professor, The Psychotherapy Department,
Medical University of Łódź
(Poland)

S 23-4

Abstract → Next

" Healing Beyond the Mind: The Impact of Psychotherapy on the Immunological Profile"

CURRICULUM VITAE

Małgorzata Gałecka, Ph.D., is an Assistant Professor and Head of the Psychotherapy Department at the Medical University of Łódź, Poland. She is a clinical psychologist and certified psychotherapist with extensive psychotherapeutic and academic experience at the Medical University of Łódź since 2010. Since 2018, her research has focused on psychoneuroimmunology, depressive disorders, and integrated psychiatric and psychotherapeutic care. Since 2021, she has contributed to international research and publications investigating the inflammatory, neurodevelopmental, and molecular mechanisms of depression. Her selected PubMed- and SCI-indexed publications include studies on inflammatory and anti-inflammatory profiles in major depressive disorders, particularly the roles of IL-17, IL-21, IL-23, IL-35, and Foxp3, as well as research exploring the neurodevelopmental theory of depression.

←Back to Speaker CV

Healing Beyond the Mind: The Impact of Psychotherapy on the Immunological Profile

Małgorzata Gałecka, Ph.D.

Assistant Professor, The Psychotherapy Department, Medical University of Łódź

Abstract

While psychiatric and psychotherapeutic interventions are historically compartmentalized, contemporary psychoneuroimmunology advocates for an integrated treatment model to achieve systemic biological healing. This presentation explores the synergistic impact of a combined psychiatric and psychodynamic approach on the patient's immunological profile. Chronic interpersonal stress, developmental adversity, and unresolved unconscious conflicts persistently prime the immune system, leading to hyperactivation of the hypothalamic-pituitary-adrenal (HPA) axis and a chronic pro-inflammatory state that impairs neuroplasticity. Within an integrated framework, targeted psychiatric treatments serve to acutely downregulate central and peripheral inflammatory cascades, thereby opening a crucial window of biological neuroplasticity. Concurrently, long-term psychodynamic psychotherapy leverages this receptive state; through the therapeutic alliance, affective processing, and the resolution of maladaptive defence mechanisms, it facilitates deep structural personality changes. Clinical insights indicate that this dual approach yields superior, lasting immunoregulation compared to monotherapies, shifting the cytokine balance away from a chronic pro-inflammatory fingerprint. Ultimately, this presentation underscores that integrating psychiatric stabilization with psychodynamic restructuring is essential for breaking the cycle of neuroinflammation and achieving comprehensive, somatic-psychic remission.

← Back to Index

S24. AI in Psychiatry: From Research to Clinic

Chair: Taishiro Kishimoro, M.D., Ph.D.

Time: 15:00 - 16:00 (60 min)

Time	Speakers	Title
S 24- 1	Taishiro Kishimoro, M.D., Ph.D. (Japan)	Artificial Intelligence in Society and Psychiatric Care: A Narrative Review of Opportunities and Challenges
S 24- 2	**Kuo-Ching Liang, Ph.D. (Japan)**	Wearable-Derived Physiological Markers for Depression Screening and Severity Estimation
S 24- 3	**Hironobu Nakamura, M.D., Ph.D. (Japan)**	Language as a Clinical Interface: NLP-Based Assessment, Illness Monitoring, and Responsible AI in Schizophrenia
16:00 - 16:40		PK4.

←Back to Session 24 line up



SESSION CHAIR

Taishiro Kishimoro, M.D., Ph.D.

Professor, Center for the Promotion of Interdisciplinary Research in Medicine and
Life Science, Keio University School of Medicine and Graduate School of Medicine
(Japan)

S - 24

15:00 - 16:00

S24: Psychiatry in the Age of AI: From Research to Clinical Implementation

" Summary of Session"

Artificial intelligence (AI) has advanced at an unprecedented pace in recent years. Beyond traditional discriminative AI for classification and prediction, the emergence of generative AI, particularly large language models (LLMs), is transforming industries and reshaping the way services are delivered across society. Psychiatry is no exception. AI is opening new possibilities in a wide range of clinical applications, including diagnostic support, symptom assessment, disease monitoring, treatment decision support, and patient communication. This symposium, "Psychiatry in the Age of AI: From Research to Clinical Implementation" will provide an overview of recent advances in AI for psychiatry from three complementary perspectives. Dr. Kishimoto will provide an overview of the current landscape of AI, discussing its opportunities and challenges from broader societal perspectives to psychiatric research and clinical practice. Dr. Liang will present the development of a wristband-based wearable device to support depression care through AI-assisted screening and symptom severity assessment. Dr. Nakamura will introduce recent work applying natural language processing (NLP) to identify linguistic features associated with schizophrenia. By bringing together these complementary approaches, this symposium aims to highlight both the opportunities and the challenges of AI in psychiatry and to stimulate discussion on how AI can be responsibly translated into clinical practice while maintaining high-quality, patient-centered care.

←Back to Session 24 line up



SPEAKER

Taishiro Kishimoro, M.D., Ph.D.

Professor, Center for the Promotion of Interdisciplinary Research in Medicine and Life Science, Keio University School of Medicine and Graduate School of Medicine
(Japan)

S 24-1 Abstract → Next

"Artificial Intelligence in Society and Psychiatric Care: A Narrative Review of Opportunities and Challenges"

CURRICULUM VITAE

Prof. Taishiro Kishimoto, Ph.D., received his Ph.D. from Keio University, School of Medicine. He is currently a Professor at the Center for the Promotion of Interdisciplinary Research in Medicine and Life Science, Keio University School of Medicine and Graduate School of Medicine, Tokyo, Japan (April 2025 – Present). He was also a Professor at Hills Joint Research Laboratory for Future Preventive Medicine and Wellness, Keio University School of Medicine, Tokyo, Japan (June 2021 – March 2025). He serves as Director of the Division of Industry-Academia Collaboration and Innovation, Center for the Promotion of Medical Science Research Collaboration, Graduate School. He received the Keio Medical Science Prize Rising Star Award in January 2020 and the Psychiatry and Clinical Neurosciences Folia Award in January 2025.

←Back to Speaker CV

Artificial Intelligence in Society and Psychiatric Care: A Narrative Review of Opportunities and Challenges

Taishiro Kishimoro, M.D., Ph.D.
Professor, Center for the Promotion of Interdisciplinary Research in Medicine and Life Science, Keio University School of Medicine and Graduate School of Medicine

Abstract

Background:

Artificial intelligence (AI) is rapidly transforming both society and psychiatric care, creating unprecedented opportunities as well as new challenges. To promote the responsible development and clinical implementation of AI in psychiatry, a comprehensive understanding of its societal impact and clinical applications is essential.

Methods:

A narrative review.

Results:

The opportunities and challenges of AI were reviewed from psychological and psychiatric perspectives, focusing on two domains: society and psychiatric care. In society, AI and digital technologies are exerting an increasingly profound influence on human cognition, behavior, and communication. Even before the emergence of generative AI, echo chambers and filter bubbles had substantially influenced the information people were exposed to and the ways in which they made decisions. With the rapid adoption of generative AI, the processes of human communication and thought formation may undergo even more profound changes. Concerns have already been raised regarding harmful applications, including online gambling, as well as excessive trust in and psychological dependency on AI. At the same time, AI is increasingly used, particularly by young people, for psychological consultation and emotional support. In psychiatric care, AI has been applied to diagnostic support, symptom assessment, disease monitoring, relapse prediction, treatment decision support, and prevention. These technologies may enable more objective assessment, continuous real-world monitoring, and personalized care. However, ethical and societal issues, including fairness, privacy, explainability, accountability, and the patient–clinician relationship, remain major challenges.

Conclusions:

AI has the potential to transform psychiatric care, but responsible implementation requires rigorous evaluation, appropriate governance, and a human-centered approach.

←Back to Session 24 line up



SPEAKER

Kuo-Ching Liang, Ph.D.

Project Senior Assistant Professor, Center for the Promotion of Interdisciplinary Research in Medicine and Life Science, Keio University School of Medicine and Graduate School of Medicine
(Japan)

S 24-2

Abstract → Next

" Wearable-Derived Physiological Markers for Depression Screening and Severity Estimation"

CURRICULUM VITAE

←Back to Speaker CV

Wearable-Derived Physiological Markers for Depression Screening and Severity Estimation

Kuo-Ching Liang, Ph.D.

Project Senior Assistant Professor, Center for the Promotion of Interdisciplinary Research in Medicine and Life Science, Keio University School of Medicine and Graduate School of Medicine

Abstract

Background: Depression is common, frequently under-detected, and typically assessed through episodic clinician interviews that capture only a snapshot of a patient's state. Consumer-grade wearables continuously record physiological signals such as heart rate variability, activity, sleep, and skin temperature, that plausibly reflect the autonomic and behavioral disturbances of depression, offering a route to objective, passive, and repeatable assessment. We investigated whether features derived from wrist-worn wearable data can (1) screen for the presence of depression and (2) estimate its severity, benchmarked against gold- standard clinician ratings. Methods: In a multi-site clinical study, participants wore one of two wearable devices that captured pulse-to-pulse interval (PPI) and multi-channel vital-sign data. From these signals we engineered roughly 300 features spanning heart-rate-variability metrics (time-domain, frequency-domain etc.) and vital-sign distributions (steps, energy expenditure etc.), and supplemented with selected self-report items. Two prediction tasks were defined against clinician assessment: a binary screening task (depressed vs. non-depressed) and a severity task predicting the 17-item Hamilton Depression Rating Scale (HAM-D) total score. Gradient-boosted tree models (XGBoost) were trained with randomized hyperparameter search under subject-level cross-validation, ensuring no participant appeared in both training and test folds. Screening performance was evaluated by accuracy, sensitivity, specificity, and predictive values; severity by mean absolute error and Pearson correlation. We additionally examined feature importance and performed subgroup analyses by geographic region and by referring-clinician type (primary care vs. psychiatry). Results: Preliminary results show that for screening, the model achieved sensitivity of ≥90.0%, specificity ≥80.0%, accuracy ≥85%. For severity estimation, predicted HAM-D scores tracked clinician ratings with a mean absolute error of ≤4.5 and correlation of ≥0.65. Conclusions: These preliminary findings indicate that passively collected wearable physiological data carry signal relevant to both the detection and grading of depression. If validated prospectively, such models could support scalable, low- burden screening and between-visit monitoring, complementing, not replacing, clinical judgment. Ongoing work focuses on external validation, prospective evaluation, and interpretability of the physiological markers driving predictions.

←Back to Session 24 line up



SPEAKER

Hironobu Nakamura, M.D., Ph.D.

Assistant Professor, Department of Psychiatry and Behavioral Sciences, Science Tokyo Hospital, Institute of Science Tokyo
(Japan)

S 24-3

Abstract → Next

" Language as a Clinical Interface: NLP-Based Assessment, Illness Monitoring, and Responsible AI in Schizophrenia"

CURRICULUM VITAE

←Back to Speaker CV

Language as a Clinical Interface: NLP-Based Assessment, Illness Monitoring, and Responsible AI in Schizophrenia

Hironobu Nakamura, M.D., Ph.D.

Assistant Professor, Department of Psychiatry and Behavioral Sciences, Science Tokyo Hospital, Institute of Science Tokyo

Abstract

Background. Language is a central interface between subjective experience, psychopathology, and clinical assessment. Recent advances in natural language processing (NLP) have enabled quantitative assessment of speech abnormalities in schizophrenia, with potential applications in symptom evaluation, illness-course monitoring, and treatment-response assessment. However, the clinical translation of NLP also raises important ethical and practical concerns. Method. This presentation builds on our recent multilevel NLP study of Japanese speech in schizophrenia. Speech samples from 104 patients with schizophrenia and 101 healthy controls were collected across free conversation, storytelling, and picture description. Using GiNZA, Word2Vec, TF-IDF, and SentenceBERT, we extracted 76 morphosyntactic, semantic, and discourse features. Factor analysis, generalized estimating equation modeling, and validation procedures were used to identify interpretable linguistic markers. I will extend these findings toward future clinical applications and discuss the risks of AI implementation in psychiatry. Results. Our work identified a tripartite linguistic phenotype of schizophrenia: diminished morphosyntactic explicitness, semantic narrowing, and reduced modification-based contextual modulation. These features showed strong diagnostic discrimination in free conversation and cross-task relevance, suggesting that spontaneous speech may provide clinically meaningful indices for evaluating severity, functional impairment, treatment response, and longitudinal illness trajectories. Conclusion. NLP may help psychiatry move from episodic, clinician-rated assessment toward more continuous and ecologically valid monitoring of language and mental states. Nevertheless, responsible implementation requires safeguards against privacy invasion, algorithmic bias, model opacity, overdiagnosis, and excessive reliance on automated judgment. NLP-based AI should augment—not replace—clinical understanding, therapeutic relationships, and human responsibility.

←Back to day 2 schedule



Plenary Keynote 4 - Speaker

Christoph U. Correll, M.D.

Professor, Department of Psychiatry, Zucker School of Medicine at Hofstra/Northwell

(USA)

PK- 4 | 16:00 - 16:40 | Abstract → Next

"Harnessing Artificial Intelligence to Improve Clinical Outcomes in Mental Health: The Role of Large Language Models and Clinician Decision Support Systems"

CURRICULUM VITAE

Prof. Correll has authored >1,200 articles that have been cited >110,000 times and received over 40 research awards. In 05/2026, his h-index was 165. Since 2017, Dr. Correll has consistently been the number one most cited scientist worldwide in >10 areas, including "central nervous system agents", "psychotropic drugs", "schizophrenia", "schizophrenia spectrum and other psychotic disorders", "antipsychotics", "delayed action preparations" and "weight gain".

←Back to Speaker CV

Harnessing Artificial Intelligence to Improve Clinical Outcomes in Mental Health: The Role of Large Language Models and Clinician Decision Support Systems

Christoph U. Correll, M.D.
Professor, Department of Psychiatry, Zucker School of Medicine at Hofstra/Northwell

Abstract

Artificial intelligence (AI) is rapidly transforming medicine, and mental health care may be among the fields with the greatest unmet need—and the greatest opportunity for impact. Despite advances in psychopharmacology and psychotherapy, psychiatric care continues to face major challenges, including diagnostic heterogeneity, limited access to specialists, fragmented longitudinal data, insufficient measurement-based care, variable treatment quality, and poor prediction of outcomes. At the same time, clinicians are increasingly overwhelmed by documentation burden, expanding knowledge requirements, and growing complexity of treatment choices. This keynote lecture will examine how recent developments in generative AI, particularly large language models (LLMs), together with clinician decision support systems (CDSS), may fundamentally reshape psychiatric assessment, treatment planning, monitoring, and delivery of care. The presentation will review the current state of AI applications in mental health, ranging from automated documentation, clinical summarization, and conversational agents to predictive analytics, digital phenotyping, precision treatment selection, and augmentation of shared decision-making. Particular attention will be paid to how LLMs can synthesize multimodal information—including clinical interviews, electronic health records, patient-reported outcomes, speech and language patterns, wearable device data, and longitudinal trajectories—to support more personalized and measurement-based psychiatric care. The lecture will also discuss emerging evidence regarding AI-assisted identification of suicide risk, relapse prediction, treatment resistance, adverse effect monitoring, and early intervention strategies in severe mental illnesses such as schizophrenia, bipolar disorder, and major depressive disorder. Real- world examples of clinician-facing AI systems and integrated digital platforms will illustrate both the promise and current limitations of these technologies. Importantly, the talk will emphasize that AI should not be viewed as a replacement for clinicians, but rather as a force multiplier that can enhance clinical reasoning, efficiency, consistency, and patient engagement while preserving the central therapeutic relationship.

Awards

← Back to Index

Overseas Travel Awards (OTA)

Awardees

- **PP007 Yuzuki Sugimoto (Japan)**
  Distinct Anterior and Posterior Skull Bone Marrow Compartments Exhibit Differential Sensitivity to Commensal Microbiota Depletion

5-min Poster Blitz Awards

Awardees

- **PP003 Szu-Han Chen (Taiwan)**
  Effects of Omega-3 Polyunsaturated Fatty Acids on Metabolic Parameters in Patients Treated With Atypical Antipsychotics: An Exploratory Meta-analysis of Randomized Controlled Trials
- **OP004 Sanae Hasegawa-Ishii (Japan)**
  Nasal, but not Gut, Microbiota are Associated With Hippocampal Neuroimmune Alterations in a Mouse Model of Eosinophilic Chronic Rhinosinusitis
- **5PB007 Jianjung Ying (Taiwan)**
  Blunted Nocturnal HRV Recovery, Reduced Physical Activity, and Lower Sleep Quality in Major Depressive Disorder: A Digital Phenotyping Study Using Wearable Devices in a Taiwanese Cohort
- **5PB006 Aimal Khan (Taiwan)**
  PUFA-derived Lipid Mediator and Endocannabinoid Signaling Pathways in Major Depressive Disorder: A Mendelian Randomization Study
- **5PB013 Wen-Chin Yu (Taiwan)**
  Food Consumption by Degree of Processing and Emotional Distress Among Taiwanese Adolescents and Adults

Best Poster Award

- **The Best Poster Award will be judged by the scientific committee on the posters provided and the result will be announced and the award will be presented at the Award Ceremony on November 15, 2026.**
  Announcement of Best Poster Awards: November 14, 2026

Registration

← Back to Index

Registration Information

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← Back to Index

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